

SingleRAN

LTE and NR Power Allocation Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

(TDD) Added activation and deactivation command examples of some parameters. For details, see [6.4.1.1 Data Preparation](#) and [6.4.1.2 Using MML Commands](#).

(TDD) Revised descriptions of how to check whether L/NR Inter-band Deep Coordination has taken effect on the LTE side. For details, see [6.4.2 Activation Verification](#).

(TDD) Revised descriptions of the switch for interference coordination between public and dedicated networks in [5.3.2.3 Function Impacts](#), [6.3.2.3 Function Impacts](#), and [7.3.2.3 Function Impacts](#).

(TDD) Revised the principles and networking requirements of L/NR Inter-band Deep Coordination, and deleted descriptions of the `LTE_NR_POWER_SHARING_OPT_SW` option of the `NRDUCellAlgoSwitch.DynPowerSharingSwitch` parameter. For details, see [6.1 Principles](#), [6.3.4 Networking](#), [6.4.1.1 Data Preparation](#), and [6.4.1.2 Using MML Commands](#).

Revised descriptions in this document.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 07 (2025-12-14).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added scheduling optimization for LTE and NR dynamic power sharing. For details, see: 4.1 Principles 4.3.5 Networking 4.4.1.2 Using MML Commands	Modified parameter on the LTE side: Added the LTE_NR_DYN_PWR_SHR_SCH_OPT_SW option to the EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch parameter.	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added compatibility of 16T with LTE FDD and NR Flash Dynamic Power Sharing. For details, see: 4.3.2.1 Prerequisite Functions 4.3.3 Hardware 4.3.5 Networking	None	NR FDD LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	RAT	Base Station Model
Added L/NR Inter-band Deep Coordination. For details, see 6 L/NR Inter-band Deep Coordination (Low-Frequency TDD) .	Modified parameter on the LTE side: Added the LTE_NR_DYN_PWR_SHR_ENH_SW option to the CellDynPowerSharing.DynPowerSharingSwitch parameter. Modified MML command on the LTE side: Added the LNR Inter-Band Deep Coord Status field to the DSP CELLDYNPWRSHASTATUS command output. Modified parameters on the NR side: - Added the LNR_INTER_BAND_DEEP_COORD_SW option to the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter. - Added the NR_PWR_SHARE_CAP_ENH_SW option to the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter. Modified MML command on the NR side: Added the LNR Inter-band Deep Coord Valid Status field to the DSP NRDUCELLTRPDYNPWRS HR command output.	NR TDD LTE TDD	3900 and 5900 series base stations (macro base stations)
Added incompatibility of BTS5929R with LTE FDD and NR Flash Dynamic Power Sharing. For details, see 4.3.3 Hardware .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	RAT	Base Station Model
Added a mutually exclusive relationship of L/NR Inter-band Deep Coordination with LTE TDD and NR Flash Dynamic Power Sharing. For details, see 5.3.2.2 Mutually Exclusive Functions .	None	NR TDD LTE TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship of LTE FDD and NR Flash Dynamic Power Sharing with Static Multiple Beam 2.0. For details, see 4.3.2.3 Function Impacts .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship of LTE FDD and NR Flash Dynamic Power Sharing with LTE flexible spectrum. For details, see 4.3.2.3 Function Impacts .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship of LTE FDD and NR Flash Dynamic Power Sharing with RF module fast deep dormancy. For details, see 4.2.2 Impacts .	None	LTE FDD LTE TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added impact relationships of LTE TDD and NR Flash Dynamic Power Sharing and LTE and NR multi-band power adaptive scheduling with RF module fast deep dormancy. For details, see 5.3.2.3 Function Impacts and 7.3.2.3 Function Impacts .	None	NR FDD NR TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship of LTE FDD and NR Flash Dynamic Power Sharing with port sparseness. For details, see 4.3.2.3 Function Impacts .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Editorial Changes

Revised other descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE TDD	MRFD-161236	LTE TDD and NR Flash Dynamic Power Sharing (LTE TDD)	5 LTE TDD and NR Flash Dynamic Power Sharing
NR TDD	MRFD-161266	LTE TDD and NR Flash Dynamic Power Sharing (NR)	
LTE FDD	MRFD-181225	LTE FDD and NR Flash Dynamic Power Sharing (LTE FDD)	4 LTE FDD and NR Flash Dynamic Power Sharing
NR FDD	MRFD-181265	LTE FDD and NR Flash Dynamic Power Sharing (NR FDD)	
LTE TDD	TDLCOFD-221611	L/NR Inter-band Deep Coordination (LTE TDD)	6 L/NR Inter-band Deep Coordination (Low-Frequency TDD)
NR TDD	FOFD-101225	L/NR Inter-band Deep Coordination	
LTE TDD	TDLCOFD-210611	Multi-band Power Adaptive Scheduling	7 LTE and NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)
NR TDD	FOFD-090100	Multi-band Power Adaptive Scheduling	
None	None	High static PSD ratios between LTE TDD and NR	8 High Static PSD Ratios Between LTE TDD and NR

2.4 Differences

Table 2-1 Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
LTE TDD and NR Flash Dynamic Power Sharing (NR)	Supported only in NR TDD	5 LTE TDD and NR Flash Dynamic Power Sharing
LTE FDD and NR Flash Dynamic Power Sharing (NR FDD)	Supported only in NR FDD	4 LTE FDD and NR Flash Dynamic Power Sharing

Function Name	Difference	Chapter/Section
L/NR Inter-band Deep Coordination (Low-Frequency TDD)	Supported only in NR TDD	6 L/NR Inter-band Deep Coordination (Low-Frequency TDD)
LTE and NR multi-band power adaptive scheduling (low-frequency TDD)	Supported only in NR TDD	7 LTE and NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)
High static PSD ratios between LTE TDD and NR	Supported only in NR TDD	8 High Static PSD Ratios Between LTE TDD and NR

Table 2-2 Differences between NSA and SA

Function Name	Difference	Chapter/Section
LTE TDD and NR Flash Dynamic Power Sharing (NR)	None	5 LTE TDD and NR Flash Dynamic Power Sharing
LTE FDD and NR Flash Dynamic Power Sharing (NR FDD)	None	4 LTE FDD and NR Flash Dynamic Power Sharing
L/NR Inter-band Deep Coordination (Low-Frequency TDD)	None	6 L/NR Inter-band Deep Coordination (Low-Frequency TDD)
LTE and NR multi-band power adaptive scheduling (low-frequency TDD)	None	7 LTE and NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)
High static PSD ratios between LTE TDD and NR	None	8 High Static PSD Ratios Between LTE TDD and NR

Table 2-3 Differences between high frequency bands and low frequency bands of NR

Function Name	Difference	Chapter/Section
LTE TDD and NR Flash Dynamic Power Sharing (NR)	Supported only in low frequency bands	5 LTE TDD and NR Flash Dynamic Power Sharing

Function Name	Difference	Chapter/Section
LTE FDD and NR Flash Dynamic Power Sharing (NR FDD)	Supported only in low frequency bands	4 LTE FDD and NR Flash Dynamic Power Sharing
L/NR Inter-band Deep Coordination (Low-Frequency TDD)	Supported only in low frequency bands	6 L/NR Inter-band Deep Coordination (Low-Frequency TDD)
LTE and NR multi-band power adaptive scheduling (low-frequency TDD)	Supported only in low frequency bands	7 LTE and NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)
High static PSD ratios between LTE TDD and NR	Supported only in low frequency bands	8 High Static PSD Ratios Between LTE TDD and NR

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3 Overview

LTE and NR services are bursty, and the power on the LTE or NR side, in a given instant, can be fully occupied or become idle. LTE and NR power allocation enables 1 ms-level power sharing between LTE and NR carriers that share the same multimode RF module. The unused power of the instantaneous idle carriers can be shared to or adaptively scheduled for the carriers whose bandwidth is already occupied and where further data scheduling is still required, so that the instantaneous power used by a cell exceeds the configured static power for the cell. LTE and NR power allocation is allowed based on the fact that LTE and NR carriers do not have their power fully occupied at the same time, and helps improve power efficiency.

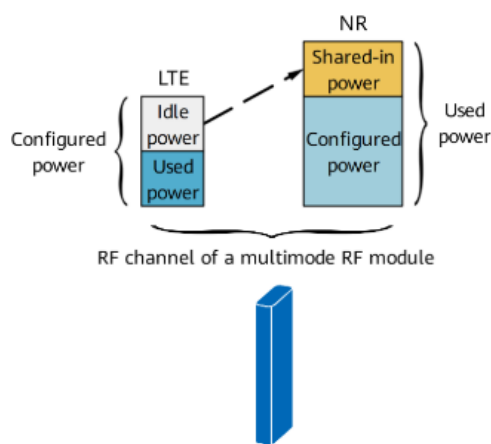
Functions of LTE and NR power allocation, as listed in [Table 3-1](#), include LTE FDD and NR Flash Dynamic Power Sharing, LTE TDD and NR Flash Dynamic Power Sharing, L/NR Inter-band Deep Coordination (Low-Frequency TDD), and LTE and NR multi-band power adaptive scheduling (low-frequency TDD).

Table 3-1 Functions of LTE and NR power allocation

Function Name	Function Description
LTE FDD and NR Flash Dynamic Power Sharing and LTE TDD and NR Flash Dynamic Power Sharing	This function enables LTE and NR carriers only on the same RF channel of an RF module to share power. It consists of LTE-to-NR power sharing and NR-to-LTE power sharing. Figure 3-1 illustrates the working principles of this function using LTE-to-NR power sharing as an example.

Function Name	Function Description
L/NR Inter-band Deep Coordination (Low-Frequency TDD)	This function enables power sharing between LTE TDD and NR carriers that share the same RF channel of an RF module . On the basis of LTE TDD and NR Flash Dynamic Power Sharing, L/NR Inter-band Deep Coordination further enables the maximum number of NR cells in each power sharing group to increase from one to two , and the maximum power sharing percentage supported by certain RF modules (such as the AAU5685md) to increase to 200% .
LTE and NR multi-band power adaptive scheduling (low-frequency TDD)	This function enables power adaptive scheduling between LTE and NR carriers that share the same RF module .
High static PSD ratios between LTE TDD and NR	This function allows operators to adjust RF configurations so that LTE cells can have a higher static power, achieving a high static power spectral density (PSD) ratio between LTE and NR.

Figure 3-1 LTE sharing power with NR



4 LTE FDD and NR Flash Dynamic Power Sharing

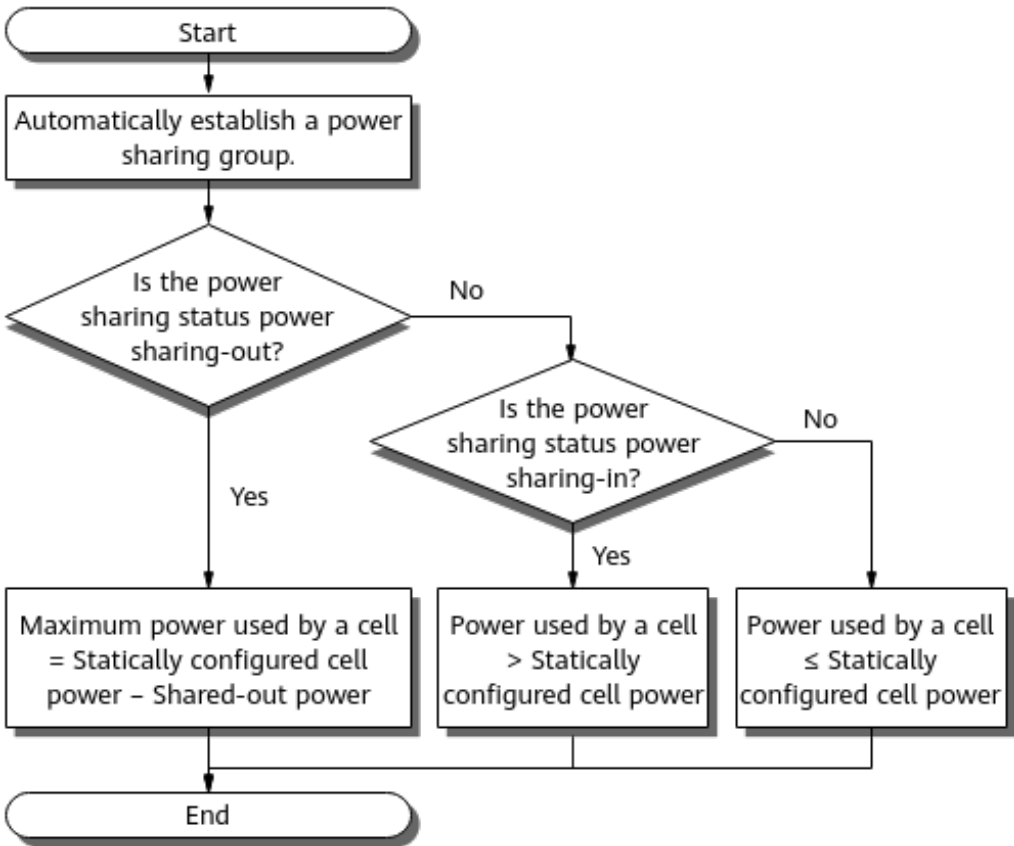
4.1 Principles

LTE FDD and NR Flash Dynamic Power Sharing takes effect only when both of the following options are selected:

- LTE: **LTE_NR_DYN_POWER_SHARING_SW** option of the **CellDynPowerSharing.DynPowerSharingSwitch** parameter
- NR: **LTE_NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch.DynPowerSharingSwitch** parameter

Figure 4-1 illustrates how LTE FDD and NR Flash Dynamic Power Sharing works.

Figure 4-1 LTE FDD and NR Flash Dynamic Power Sharing



The process is as follows:

1. Start LTE FDD and NR Flash Dynamic Power Sharing.
2. Automatically establish LTE and NR power sharing groups.
LTE and NR cells that are served by the same group of RF channels and are enabled with LTE FDD and NR Flash Dynamic Power Sharing automatically form an LTE and NR power sharing group.
Table 4-1 lists the specifications of the number of cells that support power sharing in each power sharing group in different scenarios.

Table 4-1 Specifications of the number of cells that support power sharing in each power sharing group in different scenarios

Scenario	Supported Power Sharing Cell Number Specifications	Description
2T, 4T, and 8T	Each power sharing group supports power sharing among two to nine cells.	- Each baseband processing unit supports a maximum of six LTE and NR power sharing groups. Each power sharing group can contain more than one LTE or NR cell. - Scheduling optimization for LTE and NR dynamic power sharing is controlled by the LTE_NR_DYN_PWR_SHR_SCH_OPT_SW option of the EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch parameter.
16T	- When scheduling optimization for LTE and NR dynamic power sharing is disabled, each power sharing group supports power sharing among two to nine cells. - When scheduling optimization for LTE and NR dynamic power sharing is enabled, each power sharing group supports power sharing among two to twelve cells through improvement of the efficiency of having LTE power shared.	
32T	- When scheduling optimization for LTE and NR dynamic power sharing is disabled, each power sharing group supports power sharing among two to nine cells. - When scheduling optimization for LTE and NR dynamic power sharing is enabled, each power sharing group supports power sharing among two to sixteen cells through improvement of the efficiency of sharing out LTE power.	

 NOTE

- If multiple LTE cells form an intra-RAT power sharing group but only some LTE cells are enabled with LTE FDD and NR Flash Dynamic Power Sharing, an LTE and NR power sharing group will fail to set up. Similarly, if multiple NR cells form an intra-RAT power sharing group but only some NR cells are enabled with LTE FDD and NR Flash Dynamic Power Sharing, an LTE and NR power sharing group will fail to set up.
 - If the number of cells in an LTE and NR power sharing group exceeds the maximum number of cells that are allowed to participate in power sharing, NR cannot receive shared power.
 - If multiple LTE-only power sharing groups and one NR-only power sharing group are set up on the same group of RF channels, an LTE and NR power sharing group will fail to set up. Similarly, if multiple NR-only power sharing groups and one LTE-only power sharing group are set up on the same group of RF channels, an LTE and NR power sharing group will fail to set up.
 - In RF module combination scenarios, if some LTE cells are served by one RF module and other LTE cells are served by two RF modules and if all NR cells are served by one RF module, an LTE and NR power sharing group fails to set up. Similarly, if some NR cells are served by one RF module and other NR cells are served by two RF modules and if all LTE cells are served by one RF module, an LTE and NR power sharing group fails to set up.
3. Determine the power sharing status.
- The eNodeB and gNodeB calculate the power consumption of LTE and NR cells every millisecond, and determine the cell power sharing status of each RAT for each interval. The power sharing status of each cell can be power sharing-out, power sharing-in, or no power sharing. The inter-RAT power sharing status between LTE and NR can be one of the following:
- LTE power sharing-out, and NR power sharing-in
 - LTE power sharing-in, and NR power sharing-out
 - The LTE and NR power sharing states are both power-sharing out.
 - The LTE and NR power sharing states are both power-sharing in.
- For details, see [4.1.1 Determining the Power Sharing Status](#).
4. Implement power sharing actions.
- The eNodeB and gNodeB exchange power information every 1 ms and determine the power sharing actions based on their power sharing status. The power sharing actions can be one of the following:
- LTE sharing power with NR
 - NR sharing power with LTE
 - LTE and NR sharing no power
- [Table 4-2](#) lists the mapping between the power sharing status and power sharing actions.

Table 4-2 Mapping between the power sharing status and power sharing actions

Power Sharing Status	Power Sharing Actions
LTE power sharing-out, and NR power sharing-in	LTE sharing power with NR

Power Sharing Status	Power Sharing Actions
LTE power sharing-in, and NR power sharing-out	NR sharing power with LTE
The LTE and NR power sharing states are both power-sharing out.	LTE and NR sharing no power
The LTE and NR power sharing states are both power-sharing in.	

For details about how power sharing actions are implemented, see [4.1.2 Determining Power Sharing Actions](#).

When the RF channels of an RF module serve multiple LTE or NR cells, it is recommended that LTE FDD and NR Flash Dynamic Power Sharing be enabled for all cells. Otherwise, this function cannot take effect as it is not enabled for some cells in an intra-RAT power sharing group. The principles described hereafter apply only to LTE and NR cells that are served by the RF channels of an RF module and are all enabled with this function.

4.1.1 Determining the Power Sharing Status

4.1.1.1 Determining the Power Sharing Status of LTE Cells

LTE cells calculate their power consumption and determine their power sharing status every millisecond, and notify NR cells of the power sharing status every millisecond.

The power sharing status of LTE cells in a millisecond can be power sharing-in, power sharing-out, or no power sharing.

- In the case of power sharing-in, the maximum shared-in power of a cell in a millisecond is equal to the statically configured cell power multiplied by the maximum power sharing percentage.
 - The statically configured cell power is determined by the values of the **PDSCHCfg.ReferenceSignalPwr**, **CellDlpcPdschPa.PaPcOff**, **PDSCHCfg.Pb**, and **Cell.DlBandWidth** parameters.
 - The maximum power sharing percentage is the rate of the shared power to the maximum cell power. This percentage is 100% after function activation. However, there are also special cases:
 - If LTE FDD and NR Flash Dynamic Spectrum Sharing is also enabled (by setting the **SpectrumCloud.SpectrumCloudSwitch** parameter to **LTE_NR_SPECTRUM_SHR**), the amount of power available to LTE spectrum sharing cells is less than that available to common cells, and in this case the maximum power sharing percentage is only 50%.
 - The maximum power sharing percentage may be 50% when one of the following conditions is met:

- More than three LTE or NR carriers are configured in a single frequency band of an RF module.
- For massive MIMO modules, more than five LTE and NR carriers are configured in multiple frequency bands of a multi-band RF module.
- For non-massive-MIMO modules, more than four LTE and NR carriers are configured in multiple frequency bands of a multi-band RF module.
- More than one GSM TRX is configured on an RF module.
- More than two UMTS carriers are configured in multiple frequency bands of a multi-band RF module.

Number of NR carriers = ROUNDUP(NR cell bandwidth/20 MHz). For example, if the NR cell bandwidth is 30 MHz, the number of NR carriers is two.

- The maximum power sharing percentage for RF modules is 50%.
- In the case of power sharing-out, the maximum shared-out power of a cell in a millisecond is equal to the statically configured cell power minus the required power of paging channels, control channels, and traffic bursts.

If a power sharing group contains multiple LTE cells, the power sharing status of these cells may be inconsistent. For example, the power sharing status of some cells is power sharing-in, whereas that of some other cells is power sharing-out. Under these circumstances, power sharing between LTE cells on an RF channel is preferentially triggered. [Table 4-3](#) describes how the eNodeB determines the power sharing status.

Table 4-3 Decision on power sharing status when a power sharing group contains multiple LTE cells

Condition	Power Sharing Status of LTE
Sum of the shared-out power of all LTE cells on the RF channel – Sum of the shared-in power of all LTE cells on the RF channel > 0	The power sharing status is power sharing-out. Amount of the shared-out power = Sum of the shared-out power of all LTE cells on an RF channel – Sum of the shared-in power of all LTE cells on the RF channel
Sum of the shared-out power of all LTE cells on the RF channel – Sum of the shared-in power of all LTE cells on the RF channel < 0	The power sharing status is power sharing-in. Amount of the shared-in power = Sum of the shared-in power of all LTE cells on an RF channel – Sum of the shared-out power of all LTE cells on the RF channel

Condition	Power Sharing Status of LTE
Sum of the shared-out power of all LTE cells on the RF channel – Sum of the shared-in power of all LTE cells on the RF channel = 0	The power sharing status is no inter-RAT power sharing.

5 MHz and 10 MHz LTE cells that use default channel configurations seldom share power with NR cells as these LTE cells have only few power resources left after power reservation for paging channels, control channels, and traffic bursts. LTE cells with other bandwidths can share power with NR cells as required.

4.1.1.2 Determining the Power Sharing Status of NR Cells

NR cells calculate their power consumption and determine their power sharing status every millisecond, and notify LTE cells of the power sharing status every millisecond.

The power sharing status in each millisecond can be power sharing-in, power sharing-out, or no power sharing.

- In the case of power sharing-in, the maximum shared-in power of a cell in a millisecond is equal to the statically configured cell power multiplied by the maximum power sharing percentage.
 - The statically configured cell power is specified by the **NRDUCellTrp.MaxTransmitPower** parameter.
 - The maximum power sharing percentage is the rate of the shared power to the maximum cell power. This percentage is 100% after function activation. However, there are also special cases:
 - If LTE FDD and NR Flash Dynamic Spectrum Sharing is also enabled (by setting the **SpectrumCloud.SpectrumCloudSwitch** parameter to **LTE_NR_SPECTRUM_SHR**), the amount of power available to LTE spectrum sharing cells is less than that available to common cells, and in this case the maximum power sharing percentage is only 50%.
 - The maximum power sharing percentage may be 50% when one of the following conditions is met:
 - More than three LTE or NR carriers are configured in a single frequency band of an RF module.
 - For massive MIMO modules, more than five LTE and NR carriers are configured in multiple frequency bands of a multi-band RF module.
 - For non-massive-MIMO modules, more than four LTE and NR carriers are configured in multiple frequency bands of a multi-band RF module.
 - More than one GSM TRX is configured on an RF module.
 - More than two UMTS carriers are configured in multiple frequency bands of a multi-band RF module.

Number of NR carriers = ROUNDUP(NR cell bandwidth/20 MHz). For example, if the NR cell bandwidth is 30 MHz, the number of NR carriers is two.

- The maximum power sharing percentage for RF modules is 50%.
- In the case of power sharing-out, the maximum shared-out power of a cell in a millisecond is equal to the statically configured cell power minus the power reserved for paging channels, control channels, and traffic bursts.

4.1.2 Determining Power Sharing Actions

Base stations determine the inter-RAT power sharing actions based on the power sharing status of LTE and NR cells, and then implement the power sharing actions. The inter-RAT power sharing actions can be one of the following:

- LTE sharing power with NR
- NR sharing power with LTE
- LTE and NR sharing no power

Table 4-4 describes the conditions used to determine the power sharing actions.

Table 4-4 Conditions used to determine the power sharing actions

Condition	Action
The LTE power sharing status is power sharing-out. The NR power sharing status is power sharing-in.	LTE sharing power with NR Amount of power shared between RATs (P_{share}) = min(Amount of shared-out power allowed by LTE cells, Amount of shared-in power allowed by NR cells)
The LTE power sharing status is power sharing-in. The NR power sharing status is power sharing-out.	NR sharing power with LTE Amount of power shared between RATs (P_{share}) = min(Amount of shared-in power allowed by LTE cells, Amount of shared-out power allowed by NR cells)
The LTE and NR power sharing status is both power-sharing out or power-sharing in.	LTE and NR sharing no power

The power sharing-in RAT allocates P_{share} to cells based on their power demands. In this case, the amount of power available to a cell equals the statically configured cell power plus the shared-in power.

The power sharing-out RAT apportions the P_{share} among cells based on their power sharing-out capabilities. In this case, the amount of power available to a cell equals the statically configured cell power minus the shared-out power.

4.1.3 Implementing Power Sharing Actions

LTE and NR cells allocate their respective available power to UEs based on the decision on the power sharing action. The available power includes the shared-in power.

LTE-to-NR Power Sharing Actions

In the case of LTE sharing power with NR, the remaining PDSCH power of LTE cells is dynamically shared with NR cells that have power demands and is served by the same RF channels as the LTE cells. The NR cells allocate the shared-in power to non-256QAM UEs.

When an LTE-to-NR power sharing action is determined, NR cells may have remaining RBs in the current slot.

- If PDSCH power aggregation is enabled for NR cells, the aggregated and increased power of beneficiary UEs is derived from the shared-in power and the power of remaining RBs.
- If PDSCH power aggregation is disabled for NR cells, the aggregated and increased power of beneficiary UEs is only derived from the shared-in power.

NR-to-LTE Power Sharing Actions

In the case of NR sharing power with LTE, the remaining PDSCH power of the NR cells is dynamically shared with LTE cells that have power demands and are served by the same RF channels as NR cells. LTE cells use the shared-in power for the transmission of the following RBs:

- UEs using QPSK modulation in the downlink (excluding signaling transmission and service data for MCS selection with prioritized RBs)
- UEs using all MCS indexes, except for those with MCS indexes for DL 256QAM

The LTE cells using the shared-in power may cause interference to other LTE cells, and consequently the spectral efficiency of the entire cell cluster is decreased. It is recommended that the power sharing interference control function be enabled to counteract this impact. This function is controlled by the **CellDynPowerSharing.InterferenceControlSwitch** parameter. For details, see *Dynamic Power Sharing Between LTE Carriers*.

4.2 Network Analysis

4.2.1 Benefits

LTE FDD and NR Flash Dynamic Power Sharing cannot increase the RF channel power, but it can improve power usage and increase user-perceived throughput by allowing the unused power of the instantaneous idle carriers of one-RAT to be shared with the carriers of the other RAT if the bandwidth of that RAT is already occupied and further data scheduling is still required. When this function is enabled, the sum of the power used by all the cells in a power sharing group cannot exceed the sum of their static power configurations.

It is recommended that this feature be enabled to increase the average downlink user throughput when traffic is light and instantaneously imbalanced between LTE and NR cells that share the same RF channels.

- Average downlink throughput of LTE UEs = **User Downlink Average Throughput**
- Average downlink throughput of NR UEs = **User Downlink Average Throughput (DU)**

To facilitate the calculation of benefits, it is recommended that statistics about related performance counters be collected one week before this function is enabled. For details, see [4.4.3 Network Monitoring](#).

To achieve maximal benefits, all of the following conditions must be met:

- Average downlink PRB usage $\leq 30\%$
Average downlink PRB usage of LTE cells = $\text{L.ChMeas.PRB.DL.Used.Avg} / \text{L.ChMeas.PRB.DL.Avail}$
Average downlink PRB usage of NR cells = $\text{N.PRB.DL.Used.Avg} / \text{N.PRB.DL.Avail.Avg}$
 - Power sharing between a single LTE cell and a single NR cell
 - To achieve benefits on the LTE side: The downlink PRB usage of the LTE cell is greater than or equal to 40%, and the downlink PRB usage of the NR cell is less than or equal to 20%.
 - To achieve benefits on the NR side: The downlink PRB usage of the LTE cell is less than or equal to 30%.
 - Power sharing between multiple LTE cells and multiple NR cells
To achieve benefits on both LTE and NR sides: The downlink PRB usage of both LTE and NR cells is less than or equal to 30%.
- Average downlink MCS index of LTE ≤ 15
Average downlink MCS index of LTE = $(\text{L.ChMeas.PDSCH.MCS.1} \times 1 + \dots + \text{L.ChMeas.PDSCH.MCS.28} \times 28) / (\text{L.ChMeas.PDSCH.MCS.0} + \dots + \text{L.ChMeas.PDSCH.MCS.28})$
- The PRB usage of downlink CA UEs in LTE cells is less than or equal to 60%.
PRB usage of downlink CA UEs in LTE cells = $(\text{L.ChMeas.PRB.DL.PCell.Used.Avg} + \text{L.ChMeas.PRB.DL.SCell.Used.Avg}) / \text{L.ChMeas.PRB.DL.Used.Avg}$
- The last TTI before the downlink buffer is emptied occupies less than 40% of the data transmission duration.
Proportion of the last TTI before the downlink buffer is emptied in LTE cells = $(\text{L.Thrp.Time.DL} - \text{L.Thrp.Time.DL.RmvLastTTI}) / \text{L.Thrp.Time.DL}$
Proportion of the last TTI before the downlink buffer is emptied in NR cells = $(\text{N.ThpTime.DL} - \text{N.ThpTime.DL.RmvLastSlot}) / \text{N.ThpTime.DL}$

In addition to the preceding conditions, the benefits are affected by the following factors:

- When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, there is a low probability that spectrum sharing cells share power with other cells. As a

result, the benefits provided by power sharing decrease or no benefits are provided at all.

- A lower load of the shared-out cell results in a higher probability of power sharing-out and a larger amount of shared-out power, as well as a higher probability of power sharing-in for shared-in cells and hence larger power-sharing benefits. A higher load of the shared-out cell results in the opposite effects.
- When the configured power of shared-in cells is fixed, a larger configured power of shared-out cells indicates a larger amount of shared-out power, as well as a larger amount of shared-in power for shared-in cells and hence larger power-sharing benefits. A smaller configured power of shared-out cells results in the opposite effects.
- If the amount of shared-in power is fixed, a smaller configured power of shared-in cells indicates larger power sharing benefits. A larger configured power of shared-in cells results in the opposite effects.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This feature has the following impacts on LTE and NR network performance:

- This feature increases the power consumed by the PDSCH, possibly increases the average MCS index, and reduces the downlink PRB usage.
- The impact of this feature on the downlink packet delay (LTE: **L.Traffic.DL.PktDelay.Time**; NR: **N.QoS.DL.PktDelaygNB.DU.Time**) is related to services. When the tail packet percentage (the proportion of the last TTI before the downlink buffer is emptied) is low, the downlink packet delay decreases. When the tail packet percentage is high, the downlink packet delay increases.
- When this feature is enabled, the average uplink and downlink UE throughput may decrease for cells that constantly share out power.
 - LTE: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput**.
 - NR: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput (DU)**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput (DU)**.
- Feature activation or deactivation involves the adjustment of system parameters, which causes service rate fluctuations for approximately 2 minutes and affects the cell throughput. Therefore, it is recommended that you enable this feature when the traffic is light.
- When this feature is enabled, the RF module power utilization increases. Base stations consume more power, but the total power consumption does not exceed the maximum power consumed when all cells work with a full load.
- This function requires information exchanges between cells. This may increase the CPU load of baseband processing units (**VS.BBUBoard.CPULoad.Mean**) by less than 5%.

- This function increases the instantaneous power used for PDSCH scheduling, which may increase interference on neighboring cells. If this function is enabled only on an isolated base station (with no intra-frequency cells nearby), the benefits will be small. Therefore, it is recommended that this function be enabled in contiguous intra-frequency coverage areas.
- During power sharing, LTE and NR cells estimate and coordinate the power to be allocated every millisecond. There is a possibility that power estimation is inaccurate for newly scheduled UEs after the estimation. You are not advised to enable this function when testing the uplink and downlink peak data rates of the eNodeB or gNodeB.
- In MU-MIMO pairing scenarios, this function may increase inter-UE interference, which slightly increases the average cell downlink IBLER on the LTE side. Average cell downlink IBLER on the LTE side =
$$\frac{(\text{L.Traffic.DL.SCH.QPSK.ErrTB.Ibler} + \text{L.Traffic.DL.SCH.16QAM.ErrTB.Ibler} + \text{L.Traffic.DL.SCH.64QAM.ErrTB.Ibler} + \text{L.Traffic.DL.SCH.256QAM.ErrTB.Ibler})}{(\text{L.Traffic.DL.SCH.QPSK.TB} + \text{L.Traffic.DL.SCH.16QAM.TB} + \text{L.Traffic.DL.SCH.64QAM.TB} + \text{L.Traffic.DL.SCH.256QAM.TB})}$$

4.3 Requirements

4.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE FDD	MRFD-181225	LTE FDD and NR Flash Dynamic Power Sharing (LTE FDD)	LT1SLNFDPS00	per Cell
NR	MRFD-181265	LTE FDD and NR Flash Dynamic Power Sharing (NR FDD)	NR0S0FDPSF00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

The prerequisite functions on the LTE FDD side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Power aggregation assistance	PWR_AGG_ASSIST_SW option of the CellChPwrCfg.PwrAssistanceSwitch parameter	None	Power aggregation assistance must be enabled before LTE FDD and NR Flash Dynamic Power Sharing is enabled in 4T, 8T, 16T, or 32T scenarios.
LTE FDD	BBU-RRU joint clipping	BBU_RRU_JOINT_CLIP_SW option of the ENodeBAlgoSwitch.HwCapabilityExtSwitch parameter	None	BBU-RRU joint clipping must be enabled before LTE FDD and NR Flash Dynamic Power Sharing is enabled in 4T, 8T, 16T, or 32T scenarios.
LTE FDD	Power sharing assistance	PWR_SHR_ASSIST_SW option of the CellChPwrCfg.PwrAssistanceSwitch parameter	<i>Smart 8T8R (FDD)</i>	Power sharing assistance must be enabled before LTE FDD and NR Flash Dynamic Power Sharing is enabled in 4T, 8T, 16T, or 32T scenarios.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Hybrid power sharing	LTE_HYBRID_POWER_SHARING_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	Hybrid power sharing must be enabled before LTE FDD and NR Flash Dynamic Power Sharing is enabled in 4T or 8T scenarios.
LTE FDD	Intelligent beam scheduling	JOINT_SCHEDULING_SWITCH and INTER_CARRIER_PWR_SHR_SWITCH options of the SectorSplitGroup.IntelligentBeamSchSwitch parameter	<i>Massive MIMO Enhancements (FDD)</i>	Intelligent beam scheduling must be enabled before LTE FDD and NR Flash Dynamic Power Sharing is enabled in 16T or 32T scenarios.

The prerequisite functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>

4.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions on the LTE FDD and NB-IoT sides are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Cell-level PDSCH power adjustment	CELL_LEVEL_PDSC H_POWER_ADJ_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	None
LTE FDD	Unbalanced transmit power configurations between channels ^a	eUCellSectorEqm.TxPowerOffsetAnt 0 eUCellSectorEqm.TxPowerOffsetAnt 1 eUCellSectorEqm.TxPowerOffsetAnt 2 eUCellSectorEqm.TxPowerOffsetAnt 3	None
LTE FDD	GSM and LTE Dynamic Power Sharing	CellAlgoSwitch.GL PwrShare	<i>GSM and LTE Dynamic Power Sharing</i>
LTE FDD	CSPC	CellCspcPara.Cell CspcSwitch	<i>CSPC</i>
LTE FDD	eCSPC	CellCspcPara.Celle CspcSwitch	<i>CSPC</i>
LTE FDD	PHICH inner-loop power control	PhichInnerLoopPcSwitch option of the CellAlgoSwitch.Dl PcAlgoSwitch parameter	<i>Power Control</i>
LTE FDD	Out-of-band relay	OutOfBandRelaySwitch option of the CellAlgoSwitch.RelaySwitch parameter	<i>Relay</i>

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	In-band relay	InBandRelayDeNbSwitch option of the CellAlgoSwitch.RelaySwitch parameter	<i>Relay</i>
LTE FDD	Inter-eNodeB SFN based on eNodeB coordination	Number of CellAuxEnbBind MOs ≥ 1	<i>SFN</i>
LTE FDD	Inter-eNodeB adaptive SFN/SDMA based on eNodeB coordination	Number of CellAuxEnbBind MOs ≥ 1	<i>SFN</i>
LTE FDD	Auxiliary eNodeB Cell Distributed Unit Binding Relationship	Number of CellAuxDuBind MOs ≥ 1	None
LTE FDD	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_B AND_CARR_SW option of the ContigIntraBand-Carr.ContigIntraB andCarrSw parameter	<i>Seamless Intra-Band Carrier Joining (FDD)</i>
LTE FDD	Single-carrier power sharing for 4T	SINGLE_CA_PWR_SHARE_SW_FOR_4T option of the PDSCHCfg.TxChnPowerCfgSw parameter	None
LTE FDD	Virtual 4T4R	Virtual4T4RSwitch option of the CellAlgoSwitch.E mimoSwitch parameter	<i>Virtual 4T4R (FDD)</i>
NB-IoT	NB-IoT cell	Cell.NbCellFlag	<i>NB-IoT Basics (FDD)</i>
LTE FDD	eMTC-only cell	CellEmtcAlgo.EmtcAlgoSwitch set to EMTC_ONLY_CELL_SWITCH	<i>eMTC</i>

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Voltage adjustment during simulated loading	CellSimuLoad.SimuLoadAdjVoltage set to a value other than 0	None
LTE FDD	PHICH duration	PHICHCfg.PhichDuration set to EXTENDED	<i>Physical Channel Resource Management</i>

 NOTE

a: LTE FDD and NR Flash Dynamic Power Sharing can be enabled only when **eUCellSectorEqm.TxPowerOffsetAnt0**, **eUCellSectorEqm.TxPowerOffsetAnt1**, **eUCellSectorEqm.TxPowerOffsetAnt2**, and **eUCellSectorEqm.TxPowerOffsetAnt3** are all set to **0**.

The mutually exclusive functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Pattern-free PDCCH rate matching for speed turbo UEs	ST_UE_PDCCH_NO_PATTERN_RM_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	None
NR FDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeaturesw.MultiCarrierSwitch parameter	None
NR FDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>
NR FDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None

4.3.2.3 Function Impacts

The impacted functions on the LTE FDD side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	LTE flexible spectrum	Cell.CustomizedBandWidthCfgInd set to FLEX_CFG	<i>LTE Flexible Spectrum (FDD)</i>	LTE flexible spectrum affects the amounts of shared power and received shared power in cells. As a result, LTE FDD and NR Flash Dynamic Power Sharing provides lower gains.
LTE FDD	Static Multiple Beam 2.0	SMB2_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Smart 8T8R (FDD)</i>	After a cell switches to the single-beam mode by using Static Multiple Beam 2.0, power sharing cannot be performed between this cell and other cells in the multi-beam mode.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Carrier aggregation	ENodeBAlgoSwitch.CaAlgoSwitch	<i>Carrier Aggregation</i>	The longer the duration in which LTE cells in an LTE and NR power sharing group work for CA, the lower the proportion of duration in which power sharing takes effect. Proportion of time that the function takes effect = Number of TTIs in which power sharing takes effect/Total number of TTIs in which scheduling is performed.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	On-demand TX power allocation under EME	PDSCHCfg.EmfPowerLimitSwitch	<i>On-Demand TX Power Allocation Under EME</i>	When on-demand TX power allocation under EME is enabled, power sharing estimates the number of RBs based on the standard cell bandwidth. In this case, the maximum cell power is not limited by the EME policy, and the power used by a cell can exceed the maximum power limit presented by the EME policy. Therefore, if the cell power needs to be strictly limited, it is recommended that LTE FDD and NR Flash Dynamic Power Sharing be disabled.
LTE FDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Energy saving based on flexible frequency-domain scheduling is not performed in subframes where the cell receives the shared power; it is also not performed for UEs whose PA value exceeds that configured for the cell.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LTE FDD	RF channel intelligent shutdown	CellRfShutdo wn.RfShutdo wnSwitch	<i>Energy Conservation and Emission Reduction</i>	Cells that have entered the energy saving state do not participate in LTE FDD and NR Flash Dynamic Power Sharing. These cells will attempt to participate in power sharing every ten minutes.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Intelligent power-off of carriers in the same coverage	CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE FDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Intelligent power-off of carriers in the same coverage as UMTS networks	InterRatCells hutdown.ForceShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	- A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE FDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. - When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Multi-carrier coordinated energy saving	CellShutdown . CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIER_MANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynmicPowerSharingSwitch parameter deselected does not participate in LTE FDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynmicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE FDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Dynamic carrier shutdown phase 2	CellPwrSavin gAlgo.Carrier DynShutdown nPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE FDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	GL Cell Joint Intelligent Shutdown	InterRatCells hutdown.ForceShutdownSwitch	<i>Multi-RAT Coordinated Energy Saving</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE FDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	UL Cell Joint Intelligent Shutdown	InterRatCells hutdown.For ceShutdownSwitch	<i>Multi-RAT Coordinated Energy Saving</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPower Sharing.Dyna micPowerShar ingSwitch parameter deselected does not participate in LTE FDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPower Sharing.Dyna micPowerShar ingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown . CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE FDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Low power consumption mode	CellLowPower.LowPwrSwitch	<i>Green Site</i>	Cells that have entered the energy saving state do not participate in LTE FDD and NR Flash Dynamic Power Sharing. These cells will attempt to participate in power sharing every ten minutes.
LTE FDD	UL coordinated channel shutdown	CellRfShutdown.MultiRatJointChnShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	Cells that have entered the UL coordinated channel shutdown state do not participate in LTE FDD and NR Flash Dynamic Power Sharing.
LTE FDD	LNR coordinated channel shutdown	CellRfShutdown.MultiRatJointChnShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	Cells that have entered the LNR coordinated channel shutdown state do not participate in LTE FDD and NR Flash Dynamic Power Sharing.
LTE FDD	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After a cell switches to the single-beam mode by using 8T beam mode shift for energy saving, power sharing cannot be performed between this cell and other cells in the multi-beam mode.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	The power sharing capability of LTE spectrum sharing cells is lower than that of common cells. As a result, the power sharing benefits decrease.
LTE FDD	Hybrid DSS Based on Asymmetric Bandwidth	SpectrumCloud.SpectrumCloudSwitch parameter set to LTE_NR_SPECTRUM_SHR , and the LNR_SPECTRUM_SHR_ASYM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter selected	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	
LTE FDD	PDCCH symbol number adaptation	CellPdcchAlgo.PdcchSymbolSwitch	<i>Physical Channel Resource Management</i>	When this switch is turned off, the UE_PDCCH_SYMBOL_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter is deselected, and the number of PDCCH symbols configured for the LTE cell differs from that configured for the NR cell, power sharing may fail to take effect for the LTE cell.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Downlink turbo pilot	SectorSplitGroup.TurboPilotPowerOffset	<i>Massive MIMO Enhancements (FDD)</i>	When LTE FDD and NR Flash Dynamic Power Sharing is enabled (by selecting the LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter) and downlink turbo pilot is enabled (by setting the SectorSplitGroup.TurboPilotPowerOffset parameter to a non-zero value), TM9 UEs that can use shared power are restricted to QPSK UEs. In this case, the downlink user experience may deteriorate. Therefore, it is not recommended that these two functions be enabled together.
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	Increasing the power of a cell increases PIM interference to other cells. Therefore, if power reduction in intelligent PIM interference avoidance takes effect for a cell, the cell cannot receive shared power.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUTDOWN_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	A cell with the function of channel isolation for PIM interference avoidance in effect exits the original dynamic power sharing group. When channel isolation for PIM interference avoidance is terminated, the cell joins the original dynamic power sharing group.
LTE FDD	SOC level-based low power consumption mode	IPWRM.SOCLVLADJSW and SOCLVL_BASED_LOW_POWER_SW option of the CellLowPower.LowPwrConsumptionModeExtSw parameter	<i>Green Site</i>	Cells that have entered the energy saving state cannot participate in LTE FDD and NR Flash Dynamic Power Sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After a cell switches to the two-beam mode by using 32T beam mode shift for energy saving, power sharing cannot be performed between this cell and other cells in the four-beam mode.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC</i>	In cells with eMTC introduction enabled, LTE FDD and NR Flash Dynamic Power Sharing does not take effect for UEs using CRS-based demodulation and QAM modulation in the downlink.
LTE FDD	Compact bandwidth	Cell.CustomizedBandwidthCfgInd set to CFG	<i>Compact Bandwidth (FDD)</i>	Compact bandwidth affects the amounts of shared power and received shared power in cells. As a result, LTE FDD and NR Flash Dynamic Power Sharing provides lower gains.
LTE FDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over LTE FDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE FDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Port sparseness	PORT_SPARSENESS_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When port sparseness and LTE FDD and NR Flash Dynamic Power Sharing both take effect, the gains brought by port sparseness may decrease.

The impacted functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Cells that have entered the energy saving state do not participate in LTE FDD and NR Flash Dynamic Power Sharing. These cells will attempt to participate in power sharing after exiting the energy saving state.
NR FDD	LTE FDD and NR spectrum sharing	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	The power sharing capability of NR spectrum sharing cells is lower than that of common cells. As a result, the power sharing benefits decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Hybrid DSS Based on Asymmetric Bandwidth	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter selected, and LTE_NR_FDD_SPCT_SHR_ASYM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter selected	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance</i>	Increasing the power of a cell increases PIM interference to other cells. Therefore, if power reduction in intelligent PIM interference avoidance takes effect for a cell, the cell cannot receive shared power.
NR FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUTDOWN_SW option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance</i>	LTE FDD and NR Flash Dynamic Power Sharing is terminated for a cell after channel isolation for PIM interference avoidance takes effect.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>	When this option is deselected, the LTE parameter CellPdcchAlgo.PdcchSymNumSwitch is set to OFF , and the number of PDCCH symbols configured for the LTE cell differs from that configured for the NR cell, power sharing may fail to take effect for the NR cell.
NR FDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	If there are UEs for which DL 256QAM is being used, the proportion of occasions where LTE FDD and NR Flash Dynamic Power Sharing takes effect decreases.
NR FDD	SSB power aggregation	SSB_POWER_AGG_OPT_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter	<i>Turbo Coverage (FDD)</i>	In the slots during which SSB power aggregation takes effect, the cell cannot receive shared power.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Intra-gNodeB Distributed MIMO	INTRA_GNB_DL_D_MIMO_SW option of the NRDUCellCol labServ.Multi CellMimoSwitch parameter	<i>Distributed Antenna Solutions (FDD)</i>	When coherent joint transmission in intra-gNodeB Distributed MIMO and LTE FDD and NR Flash Dynamic Power Sharing are both enabled, the power sharing gains on the NR side may decrease.
NR FDD	Inter-gNodeB Distributed MIMO	INTER_GNB_DL_D_MIMO_SW option of the NRDUCellCol labServ.Multi CellMimoSwitch parameter	<i>Distributed Antenna Solutions (FDD)</i>	When inter-gNodeB coherent joint transmission in inter-gNodeB Distributed MIMO and LTE FDD and NR Flash Dynamic Power Sharing are both enabled, the power sharing gains on the NR side may decrease.
NR FDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDIA mc.DIModulationSchemeSw parameter	<i>Modulation Schemes</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where LTE FDD and NR Flash Dynamic Power Sharing takes effect decreases.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Only low power consumption mode takes effect when both low power consumption mode and LTE FDD and NR Flash Dynamic Power Sharing are enabled and a customized backup power saving policy is used in low power consumption mode.
NR FDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over LTE FDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE FDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- LTE FDD: 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with the BBU5900, BBU5910, BBU5900A, BookBBU5901a, or BookBBU5901.

- NR: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, BBU5900A, BookBBU5901a, or BookBBU5901.

Boards

- LTE FDD main control boards: UMPT boards
- NR main control boards: UMPTE and later main control boards
- LTE FDD baseband processing units: UBBPe and later baseband processing units
- NR baseband processing units: UBBPg and later baseband processing units

RF Modules

- 4T 5000 series RF modules, such as the RRU5526et, RRU5526w, RRU5522w, RRU5522, RRU5516w, RRU5516t, RRU5516, RRU5513, RRU5513w, RRU5502, RRU5502N, RRU5502w, and RRU5501, support this function. For other models, contact Huawei engineers.
- 8T, 16T, and 32T 5000 series RF modules support this function.

4.3.4 Cells

Requirements on the LTE FDD side:

- The **SectorSplitGroup.MassiveMimoEmfPowerLimit** parameter associated with the **SectorSplitGroupExt** MO corresponding to the **SectorSplitCell** and **SectorSplitCellExt** MOs must be set to **0**.
- The **eNodeBFunction.eNodeBMode** parameter is not set to **BTRUNC_MODE**.
- If the **eUCellSectorEqm.CellBeamMode** parameter is set to **MASSIVE_MIMO_Ph2**, the **JOINT_SCHEDULING_SWITCH** and **INTER_CARRIER_PWR_SHR_SWITCH** options of the **SectorSplitGroup.IntelligentBeamSchSwitch** parameter associated with the **SectorSplitCell** and **SectorSplitCellExt** MOs must be selected.
- The **Cell.DlBandWidth** parameter is not set to **CELL_BW_N6** or **CELL_BW_N15**.
- The **CellDynPowerSharing.PowerAdjustmentMode** parameter must be set to **PDSCH_ONLY**.
- The **PDSCHCfg.CrsPwrReductionAmount** parameter must be set to **DB_0**.

Requirements on the NR side:

The **NRDUfddCellTrpBeam.BeamformingCapability** parameter cannot be set to **HORIZONTAL_ENH_MODE**.

4.3.5 Networking

The following networking requirements must be met:

- The cells are 4T, 8T, 16T, or 32T cells.
- There are no LTE carriers with a bandwidth less than 5 MHz on the RF channels serving the cells.

- The total number of LTE and NR cells served by an RF module must meet the following requirements:
 - 2T, 4T, or 8T scenarios: The total number of LTE and NR cells served by an RF module does not exceed nine.
 - 32T scenarios:
 - If the **LTE_NR_DYN_PWR_SHR_SCH_OPT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch** parameter is deselected, the total number of LTE and NR cells served by an RF module does not exceed nine.
 - If the **LTE_NR_DYN_PWR_SHR_SCH_OPT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch** parameter is selected, the total number of LTE and NR cells served by an RF module does not exceed sixteen.
 - 16T scenarios:
 - If the **LTE_NR_DYN_PWR_SHR_SCH_OPT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch** parameter is deselected, the total number of LTE and NR cells served by an RF module does not exceed nine.
 - If the **LTE_NR_DYN_PWR_SHR_SCH_OPT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch** parameter is selected, the total number of LTE and NR cells served by an RF module does not exceed twelve.

4.3.6 Others

MRFD-181225 LTE FDD and NR Flash Dynamic Power Sharing (LTE FDD) and MRFD-181265 LTE FDD and NR Flash Dynamic Power Sharing (NR FDD) must be used simultaneously.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This feature takes effect only after parameters are configured on both the eNodeB and gNodeB sides. [Table 4-5](#) and [Table 4-6](#) describe the parameters used for function activation on the eNodeB and gNodeB sides, respectively.

Table 4-5 eNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	CellDynPowerSharing.DynamicPowerSharingSwitch	Select the LTE_NR_DYN_POWER_SHARING_SW option.

Parameter Name	Parameter ID	Setting Notes
Interference Control Switch	CellDynPowerSharing.InterferenceControlSwitch	You are advised to set this parameter to ON for the local cell and its intra-frequency neighboring cells when LTE FDD and NR Flash Dynamic Power Sharing is enabled.
Network Performance Optimization Extend Sw	EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch	You are advised to select the LTE_NR_DYN_PWR_SHR_SCH_OPT_SW option when LTE FDD and NR Flash Dynamic Power Sharing is enabled in 16T and 32T scenarios.

Table 4-6 gNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	NRDUCellAlgoSwitch.DynPowerSharingSwitch	Select the LTE_NR_DYN_POWER_SHARING_SW option.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//On the eNodeB side
MOD CELLDYNPOWERSHARING: LocalCellId=0,
DynamicPowerSharingSwitch=LTE_NR_DYN_POWER_SHARING_SW-1;
MOD CELLDYNPOWERSHARING: LocalCellId=0, InterferenceControlSwitch=ON;
//Enabling scheduling optimization for LTE and NR dynamic power sharing in 16T and 32T scenarios
MOD ENODEBALGOEXTSWITCH: NetworkPrfmOptExtSwitch=LTE_NR_DYN_PWR_SHR_SCH_OPT_SW-1;

//On the gNodeB side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
DynPowerSharingSwitch=LTE_NR_DYN_POWER_SHARING_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//On the eNodeB side
MOD ENODEBALGOEXTSWITCH: NetworkPrfmOptExtSwitch=LTE_NR_DYN_PWR_SHR_SCH_OPT_SW-0;
MOD CELLDYNPOWERSHARING:
LocalCellId=0,DynamicPowerSharingSwitch=LTE_NR_DYN_POWER_SHARING_SW-0;
MOD CELLDYNPOWERSHARING: LocalCellId=0, InterferenceControlSwitch=OFF;

//On the gNodeB side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
DynPowerSharingSwitch=LTE_NR_DYN_POWER_SHARING_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Perform the following steps to check whether this feature has taken effect.

1. Check whether this feature has taken effect in LTE cells.
Run the eNodeB MML command **DSP CELLDYNPWRSHASTATUS**. If the value of **NR LTE Dynamic Power Sharing Status** is **ON**, this feature has taken effect in LTE cells.
2. Check whether this feature has taken effect in NR cells.
Run the gNodeB MML command **DSP NRDUCELLTRPDYNPWRSHR**. If the value of **LTE-NR Dyn Power Sharing Valid Status** is **Enabled**, this feature has taken effect in NR cells.
3. Monitor the values of the counters listed in the tables below on the MAE. If the counter values are greater than 0, this feature has taken effect. If the counter values are 0, this feature has not taken effect.

Table 4-7 eNodeB counters

Counter ID	Counter Name
1526746987	L.PwrShare.PwrOut.TTI.Num
1526746986	L.PwrShare.PwrIn.TTI.Num

Table 4-8 gNodeB counters

Counter ID	Counter Name
1911823028	N.PwrShare.PwrOut.Slot.Num

Counter ID	Counter Name
1911823027	N.PwrShare.PwrIn.Slot.Num

NOTE

- After LTE and NR dynamic power sharing is activated, LTE and NR cells will negotiate the configuration of dynamic power sharing. Generally, the negotiation can be completed within 5 minutes after the last activation step. Therefore, it is recommended that the activation status be queried 5 minutes after the activation procedure is complete.
- Intra-RAT power sharing is preferentially performed in an LTE and NR power sharing group. If there is still surplus power after intra-RAT power sharing, inter-RAT power sharing can be performed. The performance counters related to power sharing measure the actual power sharing-in or power sharing-out time of each cell, without distinguishing between intra-RAT and inter-RAT power sharing.
- LTE measures the number of TTIs in which a cell shares out power and the number of TTIs in which a cell receives shared power in each millisecond. NR measures the number of slots in which a cell shares out power and the number of slots in which a cell receives shared power.
- The **L.PwrShare.PwrIn.TTI.Num** counter indicates the number of TTIs within which the cell uses shared power. As power sharing uses the estimation results from the previous TTI, the estimated number of TTIs within which the cell has received shared power might be incorrect.
- The **N.PwrShare.PwrIn.Slot.Num** counter indicates the number of slots within which the cell uses shared power. As power sharing uses the estimation results from the previous millisecond, the estimated number of slots within which the cell has received shared power might be incorrect.
- When a slot of an NR cell is counted as a power sharing-in slot, the actual power used in the slot is generally greater than the statically configured cell power. However, there are also exceptions. When PDSCH power aggregation is disabled for an NR cell, the actual power used in the slot is less than the statically configured cell power if there are remaining RBs in the slot and the power of the remaining RBs is greater than the increased power of beneficiary UEs.
- When the power of a single cell is shared with one or multiple cells in an LTE and NR power sharing group, the number of TTIs in which a cell uses shared power is counted for each cell. Therefore, the number of TTIs in which a cell shares its power with other cells differs from the number of TTIs in which a cell uses shared-in power.

4.4.3 Network Monitoring

Observe the values of the following KPIs to monitor benefits:

- Average downlink throughput of LTE UEs = **User Downlink Average Throughput**
- Average downlink throughput of NR UEs = **User Downlink Average Throughput (DU)**

5 LTE TDD and NR Flash Dynamic Power Sharing

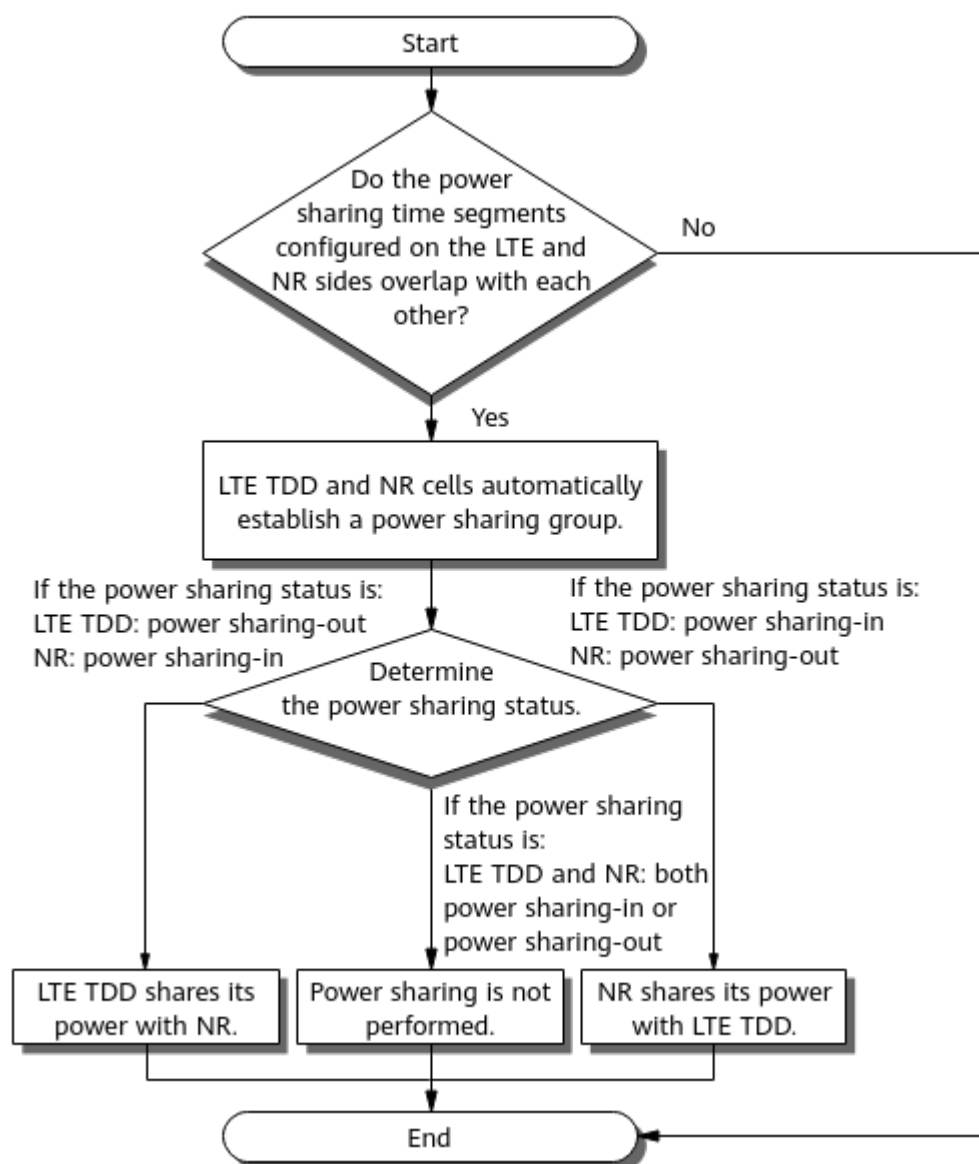
5.1 Principles

LTE TDD and NR Flash Dynamic Power Sharing takes effect only when both of the following options are selected:

- LTE: **LTE_NR_DYN_POWER_SHARING_SW** option of the **CellDynPowerSharing.DynPowerSharingSwitch** parameter
- NR: **LTE_NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch.DynPowerSharingSwitch** parameter

Figure 5-1 illustrates how LTE TDD and NR Flash Dynamic Power Sharing works.

Figure 5-1 LTE TDD and NR Flash Dynamic Power Sharing



The process is as follows:

1. Start LTE and NR dynamic power sharing.
2. Check whether the power sharing time segments configured on the LTE and NR sides overlap with each other.
 - If they do, the process goes to 3.
 - If they do not, the process ends. Dynamic Power Sharing Between LTE Carriers is performed on the LTE side. For details about this function, see *Dynamic Power Sharing Between LTE Carriers*.

The power sharing time segments on the LTE and NR sides are configured through the following parameters:

- LTE supports dynamic power sharing in a maximum of 24 time segments. Each time segment is specified by the **GlobalProcSwitch.LteNrPowerSharingTime** parameter.

- NR supports dynamic power sharing in a maximum of 24 time segments. Each time segment is specified by the **gNBDUPParam.LteNrPowerShrEnableTime** parameter.
- 3. Establish an LTE and NR power sharing group.
In the time segments that are configured for both LTE and NR, LTE and NR cells that are served by the same group of RF channels and are enabled with this function automatically form an LTE and NR power sharing group.
Each LTE or NR baseband processing unit supports a different number of LTE and NR power sharing groups, depending on its cell specifications.
 - When an LTE or NR baseband processing unit supports fewer than six cells, it supports a maximum of three LTE and NR power sharing groups.
 - When an LTE or NR baseband processing unit supports at least six cells, it supports a maximum of six LTE and NR power sharing groups.Each power sharing group can contain only one NR cell, and supports power sharing among two to six cells.
- 4. Determine the power sharing status.
The eNodeB and gNodeB calculate the power consumption of LTE and NR cells every 1 ms, and determine the cell power sharing status of each RAT for each interval. The power sharing status can be one of the following:
 - LTE power sharing-out, and NR power sharing-in
 - LTE power sharing-in, and NR power sharing-out
 - LTE and NR both power sharing-out or sharing-inFor details, see [5.1.1 Determining the Power Sharing Status](#).
- 5. Implement power sharing actions.
The eNodeB and gNodeB exchange power information every 1 ms and determine the power sharing actions based on their power sharing status. The power sharing actions can be one of the following:
 - LTE sharing power with NR
 - NR sharing power with LTE
 - LTE and NR sharing no powerFor details on how power sharing actions are implemented, see [5.1.2 Determining Power Sharing Actions](#).

When the RF channels of an RF module serve multiple LTE or NR cells, only the LTE and NR cells enabled with this function can participate in power sharing. Other cells will not affect the power sharing status. The principles described hereafter apply only to LTE and NR cells that are served by the RF channels of an RF module and are enabled with this function.

5.1.1 Determining the Power Sharing Status

5.1.1.1 Determining the Power Sharing Status of LTE Cells

LTE cells calculate their power consumption and notify NR cells of the power consumption every millisecond.

LTE cells determine their power sharing status in each TTI (1 ms). The algorithm used to determine the power sharing status is the same as that used by Dynamic

Power Sharing Between LTE Carriers. For details, see *Dynamic Power Sharing Between LTE Carriers*.

When an RF channel serves only one LTE cell, the power sharing status of the cell in a TTI (1 ms) can be power sharing-in, power sharing-out, or no power sharing.

- In the case of power sharing-in, the maximum shared-in power of a cell in a TTI is equal to the statically configured cell power multiplied by the maximum power sharing percentage.
 - The statically configured cell power is determined by the values of the **PDSCHCfg.ReferenceSignalPwr**, **CellDlpcPdschPa.PaPcOff**, **PDSCHCfg.Pb**, and **Cell.DlBandWidth** parameters.
 - The maximum power sharing percentage is the rate of the shared power to the maximum cell power.
 - If the **PWR_SHR_ASSIST_SW** option of the **CellChPwrCfg.PwrAssistanceSwitch** parameter is deselected, the maximum power sharing percentage is specified by the **CellDynPowerSharing.MaxPowerSharingRate** parameter.
 - If the **PWR_SHR_ASSIST_SW** option of the **CellChPwrCfg.PwrAssistanceSwitch** parameter is selected:
 - For 8T8R cells, the maximum power sharing percentage can reach up to 100% and is not specified by the **CellDynPowerSharing.MaxPowerSharingRate** parameter. Instead, the actual maximum power sharing percentage is determined according to the RF module capability.
 - For 32T32R and 64T64R cells, the maximum power sharing percentage is specified by the **CellDynPowerSharing.MaxPowerSharingRate** parameter.
- In the case of power sharing-out, the maximum shared-out power of a cell in a TTI (1 ms) is equal to the statically configured cell power minus the power reserved for paging channels, control channels, and traffic bursts.

If a power sharing group contains multiple LTE cells, the power sharing status of these cells may be inconsistent. For example, the power sharing status of some cells is power sharing-in, whereas that of some other cells is power sharing-out. Under these circumstances, power sharing between LTE cells is preferentially triggered. [Table 5-1](#) describes how the eNodeB determines the power sharing status.

Table 5-1 Decision on power sharing status when a power sharing group contains multiple LTE cells

Condition	Power Sharing Status of LTE
Sum of the shared-out power of all LTE cells on the RF channel – Sum of the shared-in power of all LTE cells on the RF channel > 0	The power sharing status is power sharing-out. Amount of the shared-out power = Sum of the shared-out power of all LTE cells on an RF channel – Sum of the shared-in power of all LTE cells on the RF channel
Sum of the shared-out power of all LTE cells on the RF channel – Sum of the shared-in power of all LTE cells on the RF channel < 0	The power sharing status is power sharing-in. Amount of the shared-in power = Sum of the shared-in power of all LTE cells on an RF channel – Sum of the shared-out power of all LTE cells on the RF channel
Sum of the shared-out power of all LTE cells on the RF channel – Sum of the shared-in power of all LTE cells on the RF channel = 0	The power sharing status is no power sharing.

5 MHz or 10 MHz LTE cells that use default channel configurations seldom share power with NR cells as these LTE cells have only few power resources left after power reservation for paging channels, control channels, and traffic bursts. LTE cells with other bandwidths can share power with NR cells as required.

To increase the power shared by LTE with NR, the **LNR_POWER_SHARING_RATIO_OPT_SW** option of the **CellDynPowerSharing.DynamicPowerSharingSwitch** parameter can be selected to enable LTE and NR power sharing ratio optimization. With this function, scheduling is performed in advance in the LTE cell, which then shares the remaining power after scheduling with NR in real time.

5.1.1.2 Determining the Power Sharing Status of NR Cells

NR cells calculate their power consumption and notify LTE cells of the power consumption every millisecond.

The power sharing status in each millisecond can be power sharing-in, power sharing-out, or no power sharing.

- In the case of power sharing-in, the maximum shared-in power of a cell in a millisecond is equal to the statically configured cell power multiplied by the maximum power sharing percentage.
 - The statically configured cell power is specified by the **NRDUCellTrp.MaxTransmitPower** parameter.
 - The maximum power sharing percentage is specified by the **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** parameter.

For 8T8R cells, the actual percentage is $\min\{\text{NRDUCellTrpDynPwrShr.MaxPowerSharingRate}, 100\}$. For 32T32R and 64T64R cells, the actual percentage is $\min\{\text{NRDUCellTrpDynPwrShr.MaxPowerSharingRate}, 50\}$.

- In the case of power sharing-out, the maximum shared-out power of a cell in a millisecond is equal to the statically configured cell power minus the power reserved for paging channels, control channels, and traffic bursts.

NR cells determine the power sharing status of each slot. Based on the power sharing status of the two slots in a millisecond (the descriptions in this document are based on the assumption that the NR subcarrier spacing is 30 kHz), NR cells determine the power sharing status of the millisecond.

- If the power sharing status of two slots is power sharing-out, the power sharing status of the millisecond is power sharing-out.
- If the power sharing status of at least one slot is power sharing-in, the power sharing status of the millisecond is power sharing-in.
- Under other circumstances, the power sharing status of the millisecond is no power sharing.

During power sharing, LTE and NR cells estimate and coordinate the power to be allocated every millisecond. There is a possibility that power estimation is inaccurate for newly scheduled UEs after the estimation. NR networks are lightly loaded in the initial phase of network deployment. To reduce the impact of power scheduling on NR networks, NR cells share out power only in slots that do not transmit PUSCH resource scheduling control information and the slots that do not transmit CSI reference signal (CSI-RS).

When the NR cell in the LTE and NR power sharing group meets any of the following conditions, only LTE-to-NR unidirectional power sharing is supported, considering the NR cell bandwidth, delay requirements, and sharing capability of each channel:

- The NR cell works on a 40 MHz bandwidth, and the UE-specific PDCCH occupies one symbol.
- The multi-position PDCCH scheduling function is enabled for the NR cell.
- The slot assignment of the NR cell is dual-period 8:2, and the slot structure is SS818.

Performance loss occurs in NR-to-LTE power sharing scenarios due to inaccurate power estimation for self-contained slots, in which the number of downlink symbols is less than that in downlink slots. To reduce the performance loss, the **LTE_NR_POWER_SHARING_OPT_SW** option of the **NRDUCellAlgoSwitch.DynPowerSharingSwitch** parameter can be selected to enable LTE and NR power sharing optimization. In this case, NR calculates its required power based on the actual number of downlink symbols in self-contained slots in different slot structures. This ensures sufficient power reserved for self-contained slots.

5.1.2 Determining Power Sharing Actions

Base stations determine the inter-RAT power sharing actions based on the power status of LTE and NR cells, and then implement the power sharing actions. The inter-RAT power sharing actions can be one of the following:

- LTE sharing power with NR
- NR sharing power with LTE
- LTE and NR sharing no power

Table 5-2 describes the conditions used to determine the power sharing actions.

Table 5-2 Conditions used to determine the power sharing actions

Condition	Action
The LTE power sharing status is power sharing-out. The NR power sharing status is power sharing-in.	LTE sharing power with NR Amount of power shared between RATs (P_{share}) = min(Amount of shared-out power allowed by LTE cells, Amount of shared-in power allowed by NR cells)
The LTE power sharing status is power sharing-in. The NR power sharing status is power sharing-out.	NR sharing power with LTE Amount of power shared between RATs (P_{share}) = min(Amount of shared-in power allowed by LTE cells, Amount of shared-out power allowed by NR cells)
The LTE and NR power sharing status is both power-sharing out or power-sharing in.	LTE and NR sharing no power

The power sharing-in RAT allocates the P_{share} to cells based on their power demands. In this case, the amount of power available to a cell equals the statically configured cell power plus the shared-in power.

The power sharing-out RAT apportions the P_{share} among cells based on their power sharing-out capabilities. In this case, the amount of power available to a cell equals the statically configured cell power minus the shared-out power.

5.1.3 Implementing Power Sharing Actions

LTE and NR cells allocate their respective available power to UEs based on the decision on the power sharing action. The available power includes the shared-in power.

LTE-to-NR Power Sharing Actions

In the case of LTE sharing power with NR, the remaining PDSCH and PDCCH power of LTE cells is dynamically shared with the NR cell that has power demands and is served by the same RF channels as the LTE cells. The NR cell allocates the shared-in power to non-256QAM UEs.

When an LTE-to-NR power sharing action is determined, the NR cell may have remaining RBs in the current timeslot.

- If PDSCH power aggregation is enabled for the NR cell, the aggregated and increased power of beneficiary UEs is derived from the shared-in power and the power of remaining RBs.
- If PDSCH power aggregation is disabled for the NR cell, the aggregated and increased power of beneficiary UEs is only derived from the shared-in power. To achieve optimal performance gains, it is recommended that LTE and NR dynamic power sharing be enabled together with PDSCH power aggregation. For details about PDSCH power aggregation, see *Power Control in 5G RAN Feature Documentation*.

NR-to-LTE Power Sharing Actions

In the case of NR sharing power with LTE, the remaining PDSCH and PDCCH power of the NR cell is dynamically shared with LTE cells that have power demands and are served by the same RF channels as the NR cell. LTE cells use the shared-in power for the transmission of the following RBs:

- For UEs using QPSK modulation, power is allocated to RBs for which MCS index 9 or MCS indexes lower than 9 are used.
For beamforming UEs using open-loop TM7, TM8, or TM9, power is allocated to all RBs, regardless of the MCS indexes.
- For UEs using closed-loop TM9, power is allocated to all RBs, except for the RBs when MCS indexes for DL 256QAM are used.

The LTE cells using the shared-in power may cause interference to other LTE cells, and consequently the spectral efficiency of the entire cell cluster is decreased. It is recommended that the power sharing interference control function be enabled to counteract this impact. This function is controlled by the **CellDynPowerSharing.InterferenceControlSwitch** parameter. For details, see *Dynamic Power Sharing Between LTE Carriers*.

NOTE

Only LTE-to-NR unidirectional power sharing is available in 8T high-PSD ratio scenarios.

5.2 Network Analysis

5.2.1 Benefits

LTE and NR dynamic power sharing cannot increase the RF channel power, but it can improve power usage and increase user-perceived throughput by allowing the unused power of the instantaneous idle carriers of one-RAT to be shared with the carriers of the other RAT if the bandwidth of that RAT is already occupied and further data scheduling is still required. When this function is enabled, the sum of the power used by all of the cells in a power sharing group cannot exceed the sum of their static power configurations.

It is recommended that this function be enabled when traffic is light and instantaneously imbalanced between LTE and NR cells that share the same RF channel. [Table 5-3](#) describes the function gains. To facilitate the calculation of function gains, it is recommended that statistics about related performance

counters be collected one week before this function is enabled. For details, see [5.4.3 Network Monitoring](#).

Table 5-3 LTE TDD and NR Flash Dynamic Power Sharing gains

Scenario Description	Gain on the NR Side	Gain on the LTE Side
Power sharing between a single LTE cell and a single NR cell	- The average downlink UE throughput in the NR cell increases by 2% to 8%. - The average downlink throughput of UEs in the overlapping areas in the NR cell increases by 5% to 10%.	- The average downlink UE throughput in the LTE cell increases by 0% to 4%. - The average downlink throughput of CEUs in the LTE cell increases by 0% to 5%.
Power sharing between multiple LTE cells and a single NR cell	- The average downlink UE throughput in the NR cell increases by 2% to 10%. - The average downlink throughput of UEs in the overlapping areas in the NR cell increases by 5% to 15%.	- The average downlink UE throughput in LTE cells increases by 2% to 17%. - The average downlink throughput of CEUs in LTE cells increases by 5% to 25%.

 NOTE

- In certain scenarios where NR does not support power sharing-out, only LTE-to-NR unidirectional power sharing is available, as described in [4.1.1.2 Determining the Power Sharing Status of NR Cells](#). In such scenarios, if the power sharing group includes only one LTE cell, the LTE cell does not have any gains; if the power sharing group includes multiple LTE cells, the gains of LTE cells are decreased.
- A lower load of the shared-out cell results in a higher probability of power sharing-out and a larger amount of shared-out power, as well as a higher probability of power sharing-in for shared-in cells and hence a larger power-sharing gain. A higher load of the shared-out cell results in the opposite effects.
- When the configured power of shared-in cells is fixed, a larger configured power of shared-out cells indicates a larger amount of shared-out power, as well as a larger amount of shared-in power for shared-in cells and hence a larger power-sharing gain. A smaller configured power of shared-out cells results in the opposite effects.
- If the amount of shared-in power is fixed, a smaller configured power of shared-in cells indicates a larger power sharing gain. A larger configured power of shared-in cells results in the opposite effects.

To achieve maximal benefits, all of the following conditions must be met:

- Average downlink physical resource block (PRB) usage
 - Power sharing between a single LTE cell and a single NR cell

- To achieve gains on the LTE side: The downlink PRB usage of the LTE cell is greater than or equal to 50%, and the downlink PRB usage of the NR cell is less than or equal to 30%.
- To achieve gains on the NR side: The downlink PRB usage of LTE cells is less than or equal to 40%.
- Power sharing between multiple LTE cells and a single NR cell
 - To achieve gains on both the LTE and NR sides: The downlink PRB usage of the LTE cell is less than or equal to 40%.
 - To achieve gains on the NR side: Gains are mainly derived from power sharing between LTE cells. Only few gains are derived from NR-to-LTE power sharing.

Average downlink PRB usage of LTE cells = $\frac{\text{L.ChMeas.PRB.DL.Used.Avg}}{\text{L.ChMeas.PRB.DL.Avail}}$

Average downlink PRB usage of NR cells = $\frac{\text{N.PRB.DL.Used.Avg}}{\text{N.PRB.DL.Avail.Avg}}$

- The following beneficiary UEs running at least 20% of the services in LTE cells:
 - UEs using QPSK modulation without involving signaling transmission and MCS selection with prioritized RBs

Proportion of UEs for which QPSK is used = $\frac{\text{Sum of the values of counters L.ChMeas.PDSCH.MCS.0 to L.ChMeas.PDSCH.MCS.9}}{\text{Sum of the values of counters L.ChMeas.PDSCH.MCS.0 to L.ChMeas.PDSCH.MCS.28}}$

NOTE

If the **CellDischAlgo.RBPriMcsSelectRatioThd** parameter is not set to 0, the proportion of UEs for which QPSK is used approximates to the sum of the values of counters **L.ChMeas.PDSCH.MCS.2** to **L.ChMeas.PDSCH.MCS.9** divided by the sum of the values of counters **L.ChMeas.PDSCH.MCS.0** to **L.ChMeas.PDSCH.MCS.28**.

- Beamforming UEs using open-loop TM7, TM8, or TM9 and UEs using closed-loop TM9. Such UEs do not include UEs with rank 4 or above.

Proportion of PRBs for these UEs = $\frac{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3} + \text{L.ChMeas.MIMO.PRB.CL.Rank1} + \text{L.ChMeas.MIMO.PRB.CL.Rank2} + \text{L.ChMeas.MIMO.PRB.CL.Rank3})}{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3} + \text{L.ChMeas.BF.PRB.OL.Rank4} + \text{L.ChMeas.MIMO.PRB.CL.Rank1} + \text{L.ChMeas.MIMO.PRB.CL.Rank2} + \text{L.ChMeas.MIMO.PRB.OL.Rank1} + \text{L.ChMeas.MIMO.PRB.OL.Rank2} + \text{L.ChMeas.MIMO.PRB.CL.Rank3} + \text{L.ChMeas.MIMO.PRB.CL.Rank4} + \text{L.ChMeas.MIMO.PRB.OL.Rank3} + \text{L.ChMeas.MIMO.PRB.OL.Rank4} + \text{L.ChMeas.MIMO.PRB.CL.Rank5} + \text{L.ChMeas.MIMO.PRB.CL.Rank6} + \text{L.ChMeas.MIMO.PRB.CL.Rank7} + \text{L.ChMeas.MIMO.PRB.CL.Rank8})}$

Beamforming UEs using open-loop TM7, TM8, or TM9 (there are no UEs using closed-loop TM9). Such UEs do not include UEs with rank 4 or above.

Proportion of PRBs for these UEs = $\frac{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3})}{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3})}$

**L.ChMeas.BF.PR.B.OL.Rank3 + L.ChMeas.BF.PR.B.OL.Rank4 +
 L.ChMeas.MIMO.PR.B.CL.Rank1 + L.ChMeas.MIMO.PR.B.CL.Rank2 +
 L.ChMeas.MIMO.PR.B.OL.Rank1 + L.ChMeas.MIMO.PR.B.OL.Rank2 +
 L.ChMeas.MIMO.PR.B.CL.Rank3 + L.ChMeas.MIMO.PR.B.CL.Rank4 +
 L.ChMeas.MIMO.PR.B.OL.Rank3 + L.ChMeas.MIMO.PR.B.OL.Rank4)**

- The last TTI before the downlink buffer is empty occupies less than 40% of the data transmission duration.

Proportion of the last TTI before the downlink buffer is empty in LTE cells = $(L.Thp.Time.DL - L.Thp.Time.DL.RmvLastTTI)/L.Thp.Time.DL$

Proportion of the last TTI before the downlink buffer is empty in NR cells = $(N.ThpTime.DL - N.ThpTime.DL.RmvLastSlot)/N.ThpTime.DL$

- The overlapping coverage is at most 20%.

$$\text{Overlapping coverage} = \frac{1}{k} \sum_{i=1}^k \frac{N_i^{\text{MRs when RSRP difference reaches threshold}}}{N_i^{\text{Total MRs}}}, \text{ where } k$$

indicates the total number of cells, $N_i^{\text{Total MRs}}$ is the number of valid measurement reports (MRs) for cell i , and

$N_i^{\text{MRs when RSRP difference reaches threshold}}$ is the number of MRs for cell i indicating that the reference signal received power (RSRP) difference between the serving and neighboring cells reaches the specified threshold.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This feature has the following impacts on LTE and NR network performance:

- This feature increases the power consumed by the physical downlink shared channel (PDSCH), possibly increases the average MCS index, and reduces the downlink PRB usage.
- The impact of this feature on the downlink packet delay (LTE: **L.Traffic.DL.PktDelay.Time**; NR: **N.QoS.DL.PktDelaygNBDU.Time**) is related to services. When the tail packet percentage (the proportion of the last TTI before the downlink buffer is emptied) is low, the downlink packet delay decreases. When the tail packet percentage is high, the downlink packet delay increases.
- When this feature is enabled, the average uplink and downlink UE throughput may decrease for cells that constantly share out power.
 - LTE: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput**.
 - NR: The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**, and the average uplink UE throughput is indicated by **User Uplink Average Throughput (DU)**.
- Feature activation or deactivation involves the adjustment of system parameters, which causes service rate fluctuations for approximately 2 minutes and affects the cell throughput. Therefore, it is recommended that you enable this feature when the traffic is light.

- When this feature is enabled, the RF module power utilization increases. Base stations consume more power, but the total power consumption does not exceed the maximum power consumed when all cells work with a full load.
- This feature requires information exchanges between cells. This may increase the CPU load of baseband processing units (**VS.BBUBoard.CPULoad.Mean**) by less than 5%.
- This feature increases the instantaneous power used for PDSCH, which may increase interference to neighboring cells. It is recommended that this feature be enabled in contiguous coverage areas. If the average cell throughput decreases, or if the interference to neighboring cells significantly increases after this feature is enabled, you can decrease the maximum power sharing rate to counteract the negative impacts. (Setting the maximum power sharing rate to half of its recommended value is recommended.) If the gains are not noticeable, you can increase the value of the maximum power sharing rate to achieve higher feature gains.
 - LTE: The maximum power sharing rate is specified by the **CellDynPowerSharing.MaxPowerSharingRate** parameter.
 - NR: The maximum power sharing rate is specified by the **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** parameter.
- It is recommended that this feature be disabled before the downlink peak rate tests on an eNodeB or gNodeB.

5.3 Requirements

5.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE TDD	MRFD-161236	LTE TDD and NR Flash Dynamic Power Sharing (LTE TDD)	LT1STNFDPS00	per Cell
NR	MRFD-161266	LTE TDD and NR Flash Dynamic Power Sharing (NR)	NR0STNFDPS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

The prerequisite functions on the LTE TDD side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	The LTE TDD and NR Flash Dynamic Power Sharing function requires Dynamic Power Sharing Between LTE Carriers. LTE cells enabled with Dynamic Power Sharing Between LTE Carriers automatically form an LTE power sharing group when the conditions for triggering Dynamic Power Sharing Between LTE Carriers are met. When the LTE cells included in the LTE and NR power sharing group are the same as those included in the LTE power sharing group, both of these two functions take effect. Otherwise, only Dynamic Power Sharing Between LTE Carriers takes effect, and LTE and NR dynamic power sharing does not take effect. For details about Dynamic Power Sharing Between LTE Carriers, see <i>Dynamic Power</i>

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
				<i>Sharing Between LTE Carriers.</i>

This function has no prerequisite functions on the NR side.

5.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions on the LTE TDD side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE TDD	L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>
LTE TDD	PHICH duration	PHICHCfg.PhichDuration set to EXTENDED	<i>Physical Channel Resource Management</i>
LTE TDD	PDCCH spare power allocation optimization	CellPdcchAlgo.PdcchSparePowerAllocStrage	<i>Physical Channel Resource Management</i>
LTE TDD	PDCCH power check	CellPdcchAlgo.PdcchPowerCheckSwitch	None
LTE TDD	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_BAND_CARR_SW option of the ContigIntraBandCarr.ContigIntraBandCarrSw parameter	<i>Seamless Intra-Band Carrier Joining (TDD)</i>
LTE TDD	Multi-RRU cell	Cell.MultiRruCellMode	<i>SFN</i>
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD)</i>

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE TDD	Cell-level PDSCH power adjustment	CELL_LEVEL_PDSCH_POWER_ADJ_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	None
LTE TDD	Power sharing assistance ^a	PWR_SHR_ASSIST_SW option of the CellChPwrCfg.PwrAssistanceSwitch parameter	None
LTE TDD	Soft split resource duplex ^b	None	<i>Soft Split Resource Duplex (TDD)</i>
LTE TDD	CSPC	CellCspcPara.CellCspcSwitch	<i>CSPC</i>
LTE TDD	Enhanced coordinated scheduling based power control	CellCspcPara.CelleCspcSwitch	<i>Massive MIMO Enhancements (TDD)</i>
LTE TDD	PHICH inner-loop power control	PhichInnerLoopPcSwitch option of the CellAlgoSwitch.DlPcAlgoSwitch parameter	<i>Power Control</i>
LTE TDD	PDCCH SDMA ^c	MASSIVE_MIMO_PDCCH_SDMA_SW option of the CellPdcchAlgo.SplitBeamPdcchSdmaSw parameter	<i>Massive MIMO Basics (TDD)</i>
Note: a: LTE TDD and NR Flash Dynamic Power Sharing is mutually exclusive with power sharing assistance in 32T32R and 64T64R scenarios where the SPEC_ENH_LOOSE_PSD_LIMIT option of the RRU.CUSTOMEDRFSPECSW parameter is deselected. b: The LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter must be deselected for a cell when the following conditions are met: (1) The Cell.MultiCellShareMode parameter is set to SPLIT for the cell. (2) There is an SsrdCellGroup MO where the value of the SsrdCellGroup.PrimaryLocalCellId or SsrdCellGroup.SsrdLocalCellId parameter is the same as that of the Cell.LocalCellId parameter of the cell. c: When LTE TDD and NR Flash Dynamic Power Sharing and PDCCH SDMA are both enabled, the CellPdcchAlgo.PdcchSdmaWeightType parameter cannot be set to POWER_SHARING_AMONG_3_BEAMS, POWER_SHARING_AMONG_4_BEAMS, or POWER_SHARING_AMONG_5_BEAMS in PDCCH SDMA.			

The mutually exclusive functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	L/NR Inter-band Deep Coordination cannot be enabled together with LTE TDD and NR Flash Dynamic Power Sharing.
Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	When LTE TDD and NR Flash Dynamic Power Sharing is used together with High-speed Railway Superior Experience, power estimation may be inaccurate. Therefore, the two features are mutually exclusive.
Low-frequency NR TDD	PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchingSwitch parameter	<i>Downlink Scheduling</i>	When PDCCH rate matching is enabled, the power estimation on the NR side is inaccurate, affecting LTE and NR power sharing.
Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	None

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	None
Low-frequency NR TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	None
Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	None
Low-frequency NR TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	None
Low-frequency NR TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	None

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentatation)	Description
Low-frequency NR TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	None
Low-frequency NR TDD	Slot-level power control	POWER_CONTROL_SLOT_SW option of the NRDUCellSmartPwrLock.SmartPowerLockSw parameter	None	None
Low-frequency NR TDD	Pattern-free PDCCH rate matching for speed turbo UEs	ST_UE_PDCC_H_NO_PATTERN_RM_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	None	None

5.3.2.3 Function Impacts

The impacted functions on the LTE TDD side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Scalable bandwidth	None	<i>Cell Management</i>	10 MHz LTE cells that use default channel configurations seldom share power with NR cells as these LTE cells have only few power resources left after power reservation for paging channels, control channels, and traffic bursts.
LTE TDD	DL 256QAM	CellAlgoSwitch.DL256QamAlgoSwitch	<i>Modulation Schemes</i>	When UEs for which DL 256QAM has taken effect are scheduled in a TTI on the LTE side, LTE does not use the power shared by NR.
LTE TDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDy nMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	LTE TDD and NR Flash Dynamic Power Sharing does not take effect when RF channel dynamic muting is in effect.
LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Energy saving based on flexible frequency-domain scheduling is not performed on subframes where shared power is used.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE TDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown . CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE TDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Multi-carrier coordinated energy saving	CellShutdown . CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIERMANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynmicPowerSharingSwitch parameter deselected does not participate in LTE TDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynmicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPTION_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE TDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPTION_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavin gAlgo.Carrier DynShutdown nPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE TDD and NR Flash Dynamic Power Sharing if it has entered the energy saving state. The cell will attempt to participate in power sharing every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power sharing can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Cells that have entered the energy saving state cannot participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over LTE TDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE TDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.

The impacted functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	When UEs for which DL 256QAM has taken effect are scheduled in a slot on the NR side, NR does not use the power shared by LTE.
Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When LTE TDD and NR Flash Dynamic Power Sharing and power saving scheduling are both enabled, the effects of power saving scheduling are affected. It is recommended that LTE TDD and NR Flash Dynamic Power Sharing not be enabled with this function when the traffic of LTE and NR cells is light.
Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When LTE TDD and NR Flash Dynamic Power Sharing takes effect together with energy saving based on flexible frequency-domain scheduling, the benefits of both functions decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Multi-position PDCCH scheduling	ANY_OCCASION_PDCCH_SWITCH.LowLatencySwitch parameter	None	When LTE TDD and NR Flash Dynamic Power Sharing and multi-position PDCCH scheduling are both enabled, NR cells cannot estimate the slot and power resources to be scheduled for UEs for which multi-position PDCCH scheduling takes effect. Consequently, NR cells support only power sharing-in, and do not support power sharing-out.
Low-frequency NR TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SWITCH option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When LTE TDD and NR Flash Dynamic Power Sharing is enabled together with intra-base-station DL CoMP, the power sharing benefits on the NR side may decrease.
Low-frequency NR TDD	Co-MM	INTRA_GNB_SU_COMM_SWITCH option of the NRDUCellColLabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When LTE TDD and NR Flash Dynamic Power Sharing is enabled together with Co-MM, the power sharing benefits on the NR side may decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	When LTE TDD and NR Flash Dynamic Power Sharing is enabled together with coordinated interference management, the power sharing benefits on the NR side may decrease.
Low-frequency NR TDD	Power reduction for CCUs	None	<i>Power Control</i>	When LTE TDD and NR Flash Dynamic Power Sharing is enabled together with power reduction for CCUs, the power sharing benefits on the NR side may decrease.
Low-frequency NR TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW or INTRF_COORD_ALWAYS_WORKING_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	When LTE TDD and NR Flash Dynamic Power Sharing is enabled together with interference coordination between public and dedicated networks, the power sharing benefits on the NR side may decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of timing carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of intelligent carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH,

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter is deselected for a cell, power sharing is not performed on the cell within the period specified for RF channel intelligent shutdown. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter is selected for a cell, power sharing is not performed on the cell when the cell is in the energy saving state. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Fast carrier shutdown	- Capacity-layer cells: - In intra-base-station multi-carrier scenario s: INTRA_GNB_MULTICARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter and FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingExitSwitch parameter - In inter-base-station multi-carrier scenario	<i>Energy Conservation and Emission Reduction</i>	- If the PWR_SHR_IN_PWR_SAVE_SWITCH option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of fast carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. - If the PWR_SHR_IN_PWR_SAVE_SWITCH option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
		S: INTER_GNB_MULTICAST_ARR_SD_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingSwitch</i> parameter and FAST_CARRIER_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter		messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	TTI-level channel shutdown	NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter and TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When LTE and NR Power Sharing takes effect together with TTI-level channel shutdown, the energy saving benefits of TTI-level channel shutdown decrease and downlink UE throughput may decrease.
Low-frequency NR TDD	PDCCH aggregation level reduction	PDCCH_AGG_LVL_COMPRESS_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When LTE TDD and NR Flash Dynamic Power Sharing and PDCCH aggregation level reduction are both enabled, only LTE TDD and NR Flash Dynamic Power Sharing takes effect.
Low-frequency NR TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDlMcs.DlModulationSchemeSw parameter	<i>Modulation Schemes</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where LTE TDD and NR Flash Dynamic Power Sharing and LTE and NR power sharing optimization take effect decreases.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Only low power consumption mode takes effect when both low power consumption mode and LTE TDD and NR Flash Dynamic Power Sharing are enabled and a customized backup power saving policy is used in low power consumption mode.
Low-frequency NR TDD	Maximum SSB power offset	NRDUCellTrpBeam.MaxSsbPwrOffset	<i>Power Control</i>	When LTE TDD and NR Flash Dynamic Power Sharing is in effect and the SSB power offset is set to a value greater than 3 dB, the SSB power aggregation capability decreases.
Low-frequency NR TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When LTE TDD and NR Flash Dynamic Power Sharing takes effect, the benefits of CCE partial symbol adaptation will decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of RF module fast deep dormancy arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

LTE TDD: 3900 and 5900 series base stations (macro base stations). 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

NR: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

- Main control boards: All main control boards that support 8T8R and eCPRI-capable main control boards that support 32T32R or higher support this function. For details, see section "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.
- Baseband processing units: UBBPe and later baseband processing units

RF Modules

RF modules that meet both of the following conditions support this function:

- The RF modules are 8T8R RRUs or eCPRI-capable 32T32R or higher AAUs.
- The RF modules must support LTE and NR concurrency.

5.3.4 Networking

This feature has the following networking requirements:

- LTE and NR cells must be served by the same group of RF channels.
- The LTE subframe configuration, NR slot assignment, and NR slot structure must meet the requirements listed in the table below.

SN	LTE Subframe Configuration	LTE Special Subframe Configuration	NR Slot Assignment	NR Slot Structure	Base Station Model
1	SA2	SSP7	Single-period: 8:2 Dual-period: 8:2	Single-period: SS54 Dual-period: SS84	Macro base station
2	SA2	SSP6	Single-period: 8:2 Dual-period: 8:2	Single-period: SS56 Dual-period: SS86	Macro base station
3	SA2	SSP5	Single-period: 8:2 Dual-period: 8:2	Single-period: SS518 Dual-period: SS818	Macro base station

For single period configuration, the NR cell has a 3 ms (92160 Ts) frame offset relative to the LTE cell. For dual-period configuration, the NR cell frame offset is aligned with the LTE cell frame offset. For details about the NR slot assignment and structure, see *Standards Compliance* in *5G RAN Feature Documentation*.

- LTE and NR cells support only 8T8R, 32T32R, and 64T64R.
- The LTE cells for which LTE and NR dynamic power sharing takes effect must have the Cross LTE Carriers Dynamic Power Sharing feature enabled. Otherwise, LTE and NR dynamic power sharing cannot take effect.
- When multiple NR carriers are configured on an RF channel, LTE and NR dynamic power sharing can be enabled only on one NR carrier.

5.3.5 Others

Cells

- Cells on an RF channel where there are LTE carriers of less than 5 MHz bandwidth cannot form a power sharing group.
- Dynamic power sharing can be enabled only when the cell bandwidth is greater than or equal to 5 MHz.
- An eMTC-only cell cannot be added to a power sharing group for dynamic power sharing.
- If the **Cell.TxRxMode** parameter is set to **8T8R**:
 - The **Cell.MultiRruCellFlag** parameter cannot be set to **BOOLEAN_TRUE**.
 - There must be no **SectorSplitCell** or **SectorSplitCellExt** MO configured.
- The **NRDUCCELL.DuplexMode** parameter is set to **CELL_TDD**.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

This feature takes effect only after parameters are configured on both the eNodeB and gNodeB sides. [Table 5-4](#) and [Table 5-5](#) describe the parameters used for function activation on the eNodeB and gNodeB sides, respectively.

Table 5-4 eNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	CellDynPowerSharing.DynamicPowerSharingSwitch	• Select the LTE_NR_DYN_POWER_SHARING_SW option. • Select the LTE_DYN_POWER_SHARING_SW option.
Max Power Sharing Rate	CellDynPowerSharing.MaxPowerSharingRate	The default value is recommended.
LTE and NR Power Sharing Time	GlobalProcSwitch.LteNrPowerSharingTime	Set this parameter based on the network plan.

Table 5-5 gNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	NRDUCellAlgoSwitch <i>.DynPowerSharingSwitch</i>	• Select the LTE_NR_DYN_POWER_SHARING_SW option. • Select the LTE_NR_POWER_SHARING_OPT_SW option.
Max Power Sharing Rate	NRDUCellTrpDynPwrShr.MaxPowerSharingRate	The default value is recommended.
LTE-NR Power Sharing Enable Time	gNBDUParam.LteNrPowerShrEnableTime	Set this parameter based on the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//On the eNodeB side
MOD CELLDYNPOWERSHARING: LocalCellId=0,
DynamicPowerSharingSwitch=LTE_DYN_POWER_SHARING_SW-1&LTE_NR_DYN_POWER_SHARING_SW-1,
MaxPowerSharingRate=50;
MOD GLOBALPROCSWITCH: LteNrPowerSharingTime=TIME4-1;
//On the gNodeB side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
DynPowerSharingSwitch=LTE_NR_DYN_POWER_SHARING_SW-1;
MOD NRDUCELLTRPDYNPWRSHR: NrDuCellTrpld=0, MaxPowerSharingRate=50;
ADD GNBDUPARAM: LteNrPowerShrEnableTime=TIME4-1;
```

Optimization Command Examples

```
//On the gNodeB side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
DynPowerSharingSwitch=LTE_NR_POWER_SHARING_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//On the eNodeB side
MOD CELLDYNPOWERSHARING:
```

```
LocalCellId=0,DynamicPowerSharingSwitch=LTE_DYN_POWER_SHARING_SW-0&LTE_NR_DYN_POWER_SHARING_SW-0;  
//On the gNodeB side  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
DynPowerSharingSwitch=LTE_NR_DYN_POWER_SHARING_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Perform the following steps to check whether this feature has taken effect.

1. Check whether this feature has taken effect in LTE cells.
Run the eNodeB MML command **DSP CELLDYNPWRSHASTATUS**. If the value of **NR LTE Dynamic Power Sharing Status** is **ON**, this feature has taken effect in LTE cells.
2. Check whether this feature has taken effect in NR cells.
Run the gNodeB MML command **DSP NRDUCELLTRPDYNPWRSHR**. If the value of **LTE-NR Dyn Power Sharing Valid Status** is **Enabled**, this feature has taken effect in NR cells.
3. Monitor the values of the counters listed in the tables below on the MAE. If the counter values are greater than 0, this feature has taken effect. If the counter values are 0, this feature has not taken effect.

Table 5-6 eNodeB counters

Counter ID	Counter Name
1526746987	L.PwrShare.PwrOut.TTI.Num
1526746986	L.PwrShare.PwrIn.TTI.Num

Table 5-7 gNodeB counters

Counter ID	Counter Name
1911823028	N.PwrShare.PwrOut.Slot.Num
1911823027	N.PwrShare.PwrIn.Slot.Num

 NOTE

- After LTE and NR dynamic power sharing is activated, LTE and NR cells will negotiate the configuration of dynamic power sharing. Generally, the negotiation can be completed within 5 minutes after the last activation step. Therefore, it is recommended that the activation status be queried 5 minutes after the activation procedure is complete.
- Intra-RAT power sharing is preferentially performed in an LTE and NR power sharing group. If there is still surplus power after intra-RAT power sharing, inter-RAT power sharing can be performed. The performance counters related to power sharing measure the actual power sharing-in or power sharing-out time of each cell, without distinguishing between intra-RAT and inter-RAT power sharing.
- LTE measures the number of TTIs in which a cell shares out power and the number of TTIs in which a cell receives shared power in each 1-ms TTI. NR measures the number of slots in which a cell shares out power and the number of slots in which a cell receives shared power.
- The **L.PwrShare.PwrIn.TTI.Num** counter indicates the number of TTIs within which the cell uses shared-in power. As power sharing uses the estimation results from the previous TTI, the estimated number of TTIs within which the cell has received shared power might be incorrect.
- The **N.PwrShare.PwrIn.Slot.Num** counter indicates the number of slots within which the cell uses shared-in power. As power sharing uses the estimation results from the previous millisecond, the estimated number of slots within which the cell has received shared power might be incorrect.
- When a slot of an NR cell is counted as a power sharing-in slot, the actual power used in the slot is generally greater than the statically configured cell power. However, there are also exceptions. When PDSCH power aggregation is disabled for an NR cell, the actual power used in the slot is less than the statically configured cell power if there are remaining RBs in the slot and the power of the remaining RBs is greater than the increased power of beneficiary UEs.
- When the power of a single cell is shared with one or multiple cells in an LTE and NR power sharing group, the number of TTIs in which a cell uses shared-in power is counted for each cell. Therefore, the number of TTIs in which a cell shares its power with other cells differs from the number of TTIs in which a cell uses shared-in power.

5.4.3 Network Monitoring

LTE Network Monitoring

When event A3 can be reported, determine whether this feature has taken effect based on the throughput of CEUs, which can be calculated based on the information contained in event A3 reports. The detailed steps are as follows:

- Step 1** One week before feature activation, select the **BasedA3EdgeUserSwitch** option of the **CellCounterParaGroup.CellCounterAlgoSwitch** parameter, and set **CellCounterParaGroup.EdgeUserA3Offset** to **-12**.

 NOTE

The average downlink throughput of CEUs can be measured only when the **BasedA3EdgeUserSwitch** option of the **CellCounterParaGroup.CellCounterAlgoSwitch** parameter is selected.

- Step 2** Observe the values of the following counters to monitor feature gains.
- Average downlink UE throughput = $(\text{L.Thrp.bits.DL} - \text{L.Thrp.bits.DL.LastTTI}) / \text{L.Thrp.Time.DL.RmvLastTTI}$

- Average downlink throughput of CEUs = $(L.Thrp.bits.DL.BorderUE - L.Thrp.bits.DL.LastTTI.BorderUE)/L.Thrp.Time.DL.RmvLastTTI.BorderUE$

----End

When event A3 cannot be reported, determine whether this feature has taken effect based on the cumulative distribution function (CDF) curve of the throughput.

Create the CDF curve based on values of counters listed in [Table 5-8](#). If the throughput corresponding value 5% or 10% on the CDF curve increases, this feature has taken effect.

Table 5-8 LTE counters

Counter ID	Counter Name
1526729056 to 1526729065	L.Thrp.DL.BitRate.Samp.Index0 to L.Thrp.DL.BitRate.Samp.Index9

NR Network Monitoring

- Step 1** Select the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter one week before feature activation.

 NOTE

The average downlink throughput of UEs in overlapping areas can be measured only when the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter is selected.

- Step 2** Observe the values of the following counters to monitor feature gains.
- Average downlink UE throughput = $(N.ThpVol.DL - N.ThpVol.DL.LastSlot)/N.ThpTime.DL.RmvLastSlot$
 - Average downlink throughput of UEs in overlapping areas = $(N.ThpVol.DL.CellOverlap - N.ThpVol.DL.LastSlot.CellOverlap)/N.ThpTime.DL.RmvLastSlot.CellOverlap$

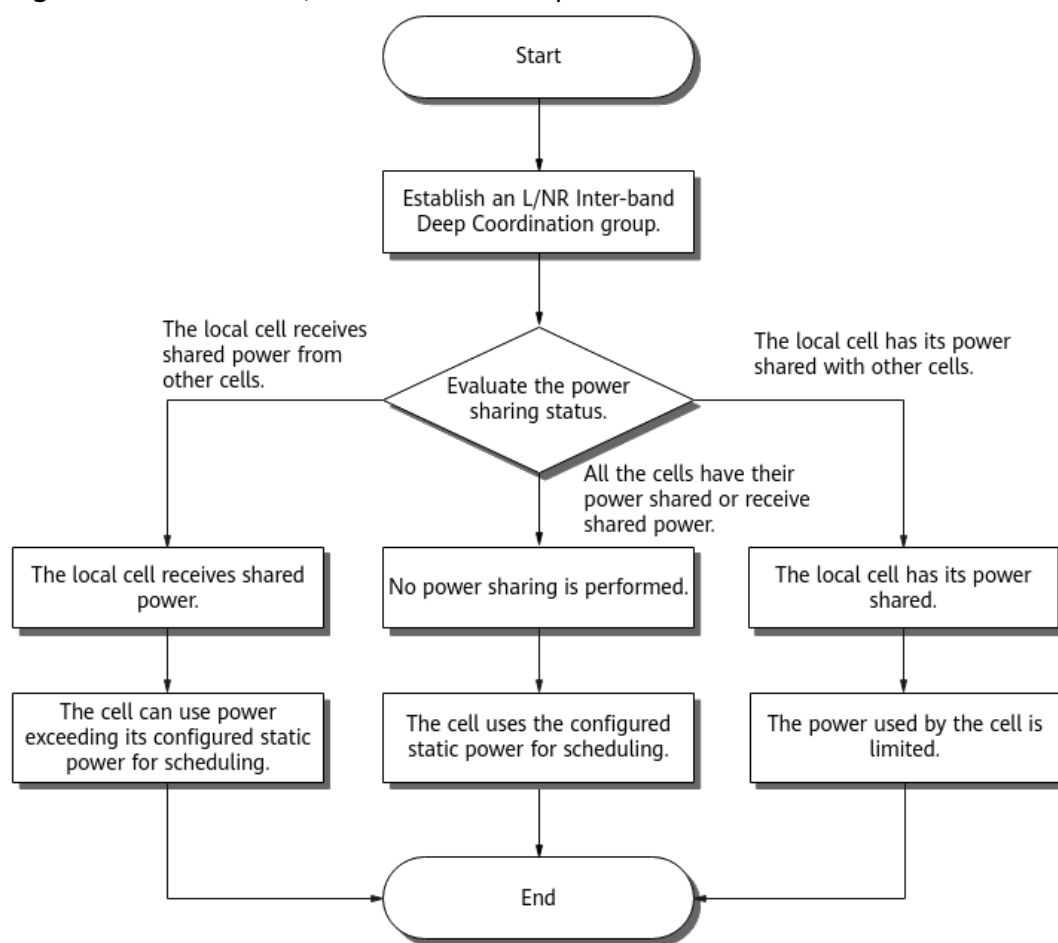
----End

6 L/NR Inter-band Deep Coordination (Low-Frequency TDD)

6.1 Principles

On the basis of LTE TDD and NR Flash Dynamic Power Sharing, L/NR Inter-band Deep Coordination further improves the power sharing percentage by enabling power sharing between several LTE cells and two NR cells sharing the same RF channel and increasing the maximum power sharing percentage supported by certain RF modules (such as the AAU5685md) to 200%. [Figure 6-1](#) shows the process.

Figure 6-1 Process of L/NR Inter-band Deep Coordination



This function requires the following options on both the LTE and NR sides to be selected:

- LTE: **LNR_INTER_BAND_DEEP_COORD_SW** option of the **CellDynPowerSharing.DynamicPowerSharingSwitch** parameter
- NR: **LNR_INTER_BAND_DEEP_COORD_SW** option of the **NRDUCellResShrCfg.DynPwrShrAlgoSwitch** parameter

Establishing an L/NR Inter-band Deep Coordination Group

LTE and NR cells that are served by the same group of RF channels and are enabled with L/NR Inter-band Deep Coordination automatically form an L/NR Inter-band Deep Coordination group.

Each LTE or NR baseband processing unit supports a different number of L/NR Inter-band Deep Coordination groups, depending on its cell specifications.

- When an LTE or NR baseband processing unit supports fewer than six cells, it supports a maximum of three L/NR Inter-band Deep Coordination groups.
- When an LTE or NR baseband processing unit supports at least six cells, it supports a maximum of six L/NR Inter-band Deep Coordination groups.

Each L/NR Inter-band Deep Coordination group allows power sharing among two to six cells and can contain a maximum of two NR cells and several LTE cells.

Determining the Power Sharing Status

The eNodeB and gNodeB check the status of each cell in a power sharing group based on the to-be-scheduled data volume estimated in advance and periodically exchange power information. They evaluate whether a cell has its power shared, receives shared power, or does not participate in power sharing.

The power sharing status of LTE cells is determined as follows:

- LTE cells in an L/NR Inter-band Deep Coordination group determine their power sharing status and notify other cells in the group of the status in each TTI (1 ms). The algorithm used to determine the power sharing status is the same as that used by Dynamic Power Sharing Between LTE Carriers. For details, see *Dynamic Power Sharing Between LTE Carriers*.
- The power sharing status in a TTI can be power sharing-in, power sharing-out, or no power sharing.
 - In the case of power sharing-in, the maximum shared-in power of a cell in a TTI is equal to the statically configured cell power multiplied by the maximum power sharing percentage.
 - The statically configured cell power is determined by the values of the **PDSCHCfg.ReferenceSignalPwr**, **CellDlpcPdschPa.PaPcOff**, **PDSCHCfg.Pb**, and **Cell.DlBandWidth** parameters.
 - The maximum power sharing percentage is the ratio of the shared power to the maximum power of the cell. [Table 6-1](#) lists the maximum power sharing percentages in different scenarios.

Table 6-1 Maximum power sharing percentages in different scenarios

Scenario	Maximum Power Sharing Percentage
PWR_SHR_ASSIST_SW option deselected	The maximum power sharing percentage is specified by the Max Power Sharing Rate parameter.
PWR_SHR_ASSIST_SW option selected	• For 8T8R cells, the maximum power sharing percentage can reach up to 100% and is not specified by the Max Power Sharing Rate parameter. Instead, the actual maximum power sharing percentage is determined according to the RF module capability. • For 32T32R and 64T64R cells, the maximum power sharing percentage is specified by the Max Power Sharing Rate parameter.
PWR_SHR_ASSIST_SW option of the CellChPwrCfg.PwrAssistanceSwitch parameter: switch for power sharing assistance Max Power Sharing Rate: CellDynPowerSharing.MaxPowerSharingRate	

- The priority of a carrier to receive allocated power in an L/NR Inter-band Deep Coordination group is specified by the **CellDynPowerSharing.MultiBandPwrAdaptSchPri** parameter. A smaller value of this parameter indicates a lower priority in an L/NR Inter-band Deep Coordination group, while a larger value indicates a higher priority.
 - In the case of power sharing-out, the maximum shared-out power of a cell in a TTI is equal to the statically configured cell power minus the power reserved for paging channels, control channels, and traffic bursts.
- 5 MHz or 10 MHz LTE cells that use default channel configurations seldom share power with other cells as these LTE cells have only few power resources left after power reservation for paging channels, control channels, and traffic bursts. LTE cells with other bandwidths can share power with NR cells as required.
- If an L/NR Inter-band Deep Coordination group consists of both LTE and NR cells, to increase the power shared by LTE with NR, the **LNR_POWER_SHARING_RATIO_OPT_SW** option of the **CellDynPowerSharing.DynamicPowerSharingSwitch** parameter can be selected to enable LTE and NR power sharing ratio optimization. With this function, scheduling is performed in advance in LTE cells, which then shares the remaining power after scheduling with NR cells in real time.

The power sharing status of NR cells is determined as follows:

- The power sharing status in each slot can be power sharing-in, power sharing-out, or no power sharing.
 - In the case of power sharing-in, the maximum shared-in power of a cell in a slot is equal to the statically configured cell power multiplied by the maximum power sharing percentage.
 - The statically configured cell power is specified by the **NRDUCellTrp.MaxTransmitPower** parameter.
 - The maximum power sharing percentage is specified by the **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** parameter.

NOTE

- When an AAU5685md is used:
 - When enhanced NR power sharing (controlled by the **NR_PWR_SHARE_CAP_ENH_SW** option of the **NRDUCellResShrCfg.DynPwrShrAlgoSwitch** parameter) is enabled, the actual percentage is $\min\{\text{NRDUCellTrpDynPwrShr.MaxPowerSharingRate}, 200, \text{RF module capability}\}$.
 - When enhanced NR power sharing is disabled, the actual percentage is $\min\{\text{NRDUCellTrpDynPwrShr.MaxPowerSharingRate}, 100, \text{RF module capability}\}$.
 - For other RF modules, the actual percentage is $\min\{\text{NRDUCellTrpDynPwrShr.MaxPowerSharingRate}, 100, \text{RF module capability}\}$.
- The priority of a carrier to receive allocated power in an L/NR Inter-band Deep Coordination group is specified by the

NRDUCellTrpDynPwrShr.MultiBandPwrAdaptSchPri parameter. A smaller value of this parameter indicates a lower priority in an L/NR Inter-band Deep Coordination group, while a larger value indicates a higher priority.

- In the case of power sharing-out, the maximum shared-out power of a cell is equal to the statically configured cell power minus the power reserved for paging channels, control channels, and traffic bursts.

When no LTE cells participate in power sharing, NR cells determine the power sharing status of each slot. When LTE cells participate in power sharing, NR inter-carrier power sharing still determines the power sharing status of each slot. The process of NR-to-LTE power sharing is as follows:

- Based on the power sharing status of the two slots in a millisecond (the descriptions in this document are based on the assumption that the NR subcarrier spacing is 30 kHz), the NR-to-LTE power sharing status of the millisecond is determined as follows:
 - If the power sharing status of two slots is power sharing-out, the power sharing status of the millisecond is power sharing-out.
 - If the power sharing status of at least one slot is power sharing-in, the power sharing status of the millisecond is power sharing-in.
 - Under other circumstances, the power sharing status of the millisecond is no power sharing.
- During power sharing, LTE and NR cells estimate and coordinate the power to be allocated every millisecond. There is a possibility that power estimation is inaccurate for newly scheduled UEs after the estimation. NR networks are lightly loaded in the initial phase of network deployment. To reduce the impact of power scheduling on NR networks, NR cells share out power only in slots that do not transmit PUSCH resource scheduling control information and the slots that do not transmit CSI-RS.
- When the NR cell in the L/NR Inter-band Deep Coordination group meets any of the following conditions, only LTE-to-NR unidirectional power sharing is supported, considering the NR cell bandwidth, delay requirements, and sharing capability of each channel:
 - The NR cell works on a 40 MHz bandwidth, and the UE-specific PDCCH occupies one symbol.
 - The multi-position PDCCH scheduling function is enabled for the NR cell.
 - The slot assignment of the NR cell is dual-period 8:2, and the slot structure is SS818.

Implementing Power Sharing Actions

Power sharing actions are performed based on the determined power sharing status.

- For cells receiving shared power, the shared power is allocated based on their power demands. In this case, the amount of power available to a cell equals the statically configured cell power plus the actually received shared power.

- For cells having their power shared, the shared power is allocated based on their power sharing-out capabilities. In this case, the amount of power available to a cell equals the statically configured cell power minus the shared-out power.
- For cells not participating in power sharing, the amount of power available to a cell equals the statically configured cell power.

LTE and NR cells allocate their respective available power to UEs based on the decision on the power sharing action. The available power includes the shared-in power.

- NR receiving shared power

If NR cells are determined to receive shared power, the remaining PDSCH and PDCCH power of the cells having their power shared is dynamically shared with NR cells that are served by the same RF channels as the cells and have power demands. The NR cells allocate the received shared power to non-256QAM UEs.

If NR cells are determined to receive shared power, they may have remaining RBs in the current slot.

- If PDSCH power aggregation is enabled for NR cells, the aggregated and increased power of beneficiary UEs is derived from the received shared power and the power of remaining RBs.
- If PDSCH power aggregation is disabled for NR cells, the aggregated and increased power of beneficiary UEs is only derived from the received shared power. To achieve optimal performance gains, it is recommended that LTE and NR dynamic power sharing be enabled together with PDSCH power aggregation. For details about PDSCH power aggregation, see *Power Control in 5G RAN Feature Documentation*.

- LTE receiving shared power

If LTE cells are determined to receive shared power, the remaining PDSCH and PDCCH power of the cells having their power shared is dynamically shared with LTE cells that are served by the same RF channels as the cells and have power demands. LTE cells use the received shared power for the transmission of the following RBs:

- For UEs using QPSK modulation, power is allocated to RBs for which MCS index 9 or MCS indexes lower than 9 are used.
- For beamforming UEs using open-loop TM7, TM8, or TM9, power is allocated to all RBs, regardless of the MCS indexes.
- For UEs using closed-loop TM9, power is allocated to all RBs, except for the RBs when MCS indexes for DL 256QAM are used.

The LTE cells using the received shared power may cause interference to other LTE cells and consequently a decrease in the spectral efficiency of the entire cell cluster. It is recommended that the interference control function be enabled to counteract this impact. This function is controlled by the **CellDynPowerSharing.InterferenceControlSwitch** parameter. For details, see *Dynamic Power Sharing Between LTE Carriers*.

NOTE

Only LTE-to-NR unidirectional power sharing is available in 8T scenarios with "high static PSD ratios between LTE TDD and NR" enabled.

6.2 Network Analysis

6.2.1 Benefits

L/NR Inter-band Deep Coordination cannot increase the RF channel power, but it can improve power usage and increase user-perceived throughput by allowing the unused power of the instantaneous idle carriers of one-RAT to be shared with the carriers of the other RAT if the bandwidth of that RAT is already occupied and further data scheduling is still required. When this function is enabled, the sum of the power used by all the cells in a power sharing group cannot exceed the sum of their static power configurations.

It is recommended that this function be enabled when traffic is light and instantaneously imbalanced between LTE and NR cells that share the same RF channel. [Table 6-2](#) describes the function gains. To facilitate the calculation of function gains, it is recommended that statistics about related performance counters be collected one week before this function is enabled. For details, see [5.4.3 Network Monitoring](#).

Table 6-2 L/NR Inter-band Deep Coordination gains

Scenario Description	Gain on the NR Side	Gain on the LTE Side
Power sharing between a single LTE cell and a single NR cell	- The average downlink UE throughput in the NR cell increases by 2% to 8%. - The average downlink throughput of UEs in the overlapping areas in the NR cell increases by 5% to 10%.	- The average downlink UE throughput in the LTE cell increases by 0% to 4%. - The average downlink throughput of CEUs in the LTE cell increases by 0% to 5%.
Power sharing between multiple LTE cells and a single NR cell	- The average downlink UE throughput in the NR cell increases by 2% to 10%. - The average downlink throughput of UEs in the overlapping areas in the NR cell increases by 5% to 15%.	- The average downlink UE throughput in LTE cells increases by 2% to 17%. - The average downlink throughput of CEUs in LTE cells increases by 5% to 25%.

Scenario Description	Gain on the NR Side	Gain on the LTE Side
Power sharing between multiple LTE cells and two NR cells	- The average downlink UE throughput in NR cells increases by 2% to 10%. - The average downlink throughput of UEs in the overlapping areas in NR cells increases by 5% to 15%.	- The average downlink UE throughput in LTE cells increases by 0% to 17%. - The average downlink throughput of CEUs in LTE cells increases by 0% to 25%.

To achieve maximal benefits, all of the following conditions must be met:

- Average downlink physical resource block (PRB) usage
 - Power sharing between a single LTE cell and a single NR cell
 - To achieve gains on the LTE side: The downlink PRB usage of the LTE cell is greater than or equal to 50%, and the downlink PRB usage of the NR cell is less than or equal to 30%.
 - To achieve gains on the NR side: The downlink PRB usage of the LTE cell is less than or equal to 40%.
 - Power sharing between multiple LTE cells and a single NR cell
 - To achieve gains on the LTE side:

Scenario 1: The downlink PRB usage of the LTE cells is greater than or equal to 40%, and the downlink PRB usage of the NR cells is less than or equal to 30%. (In this scenario, only little gains are derived from NR-to-LTE power sharing.)

Scenario 2: The downlink PRB usage of the LTE cells is less than or equal to 40%. (In this scenario, gains are mainly derived from power sharing between the LTE cells.)
 - To achieve gains on the NR side: The downlink PRB usage of the LTE cells is less than or equal to 40%. (In this scenario, gains are mainly derived from power shared from the LTE cells.)
 - Power sharing between multiple LTE cells and two NR cells
 - To achieve gains on the LTE side:

Scenario 1: The downlink PRB usage of the LTE cells is greater than or equal to 40%, and the downlink PRB usage of the NR cells is less than or equal to 30%. (In this scenario, only little gains are derived from NR-to-LTE power sharing.)

Scenario 2: The downlink PRB usage of the LTE cells is less than or equal to 40%. (In this scenario, gains are mainly derived from power sharing between the LTE cells.)
 - To achieve gains on the NR side:

Scenario 1: The downlink PRB usage of the LTE cells is less than or equal to 40%. (In this scenario, gains are mainly derived from power shared from the LTE cells.)

Scenario 2: The downlink PRB usage of the LTE cells is greater than 40%, and the downlink PRB usage of the NR cells is less than or equal to 30%. (In this scenario, gains are mainly derived from power sharing between NR cells.)

Average downlink PRB usage of LTE cells = $\frac{\text{L.ChMeas.PRB.DL.Used.Avg}}{\text{L.ChMeas.PRB.DL.Avail}}$

Average downlink PRB usage of NR cells = $\frac{\text{N.PRB.DL.Used.Avg}}{\text{N.PRB.DL.Avail.Avg}}$

- The following beneficiary UEs running at least 20% of the services in LTE cells:
 - UEs using QPSK modulation without involving signaling transmission and MCS selection with prioritized RBs

Proportion of UEs for which QPSK is used = $\frac{\text{Sum of the values of counters L.ChMeas.PDSCH.MCS.0 to L.ChMeas.PDSCH.MCS.9}}{\text{Sum of the values of counters L.ChMeas.PDSCH.MCS.0 to L.ChMeas.PDSCH.MCS.28}}$

 NOTE

If the *CellDlschAlgo.RBPrMcsSelectRatioThd* parameter is not set to 0, the proportion of UEs for which QPSK is used approximates to the sum of the values of counters *L.ChMeas.PDSCH.MCS.2* to *L.ChMeas.PDSCH.MCS.9* divided by the sum of the values of counters *L.ChMeas.PDSCH.MCS.0* to *L.ChMeas.PDSCH.MCS.28*.

- Beamforming UEs using open-loop TM7, TM8, or TM9 and UEs using closed-loop TM9 Such UEs do not include UEs with rank 4 or above.

Proportion of PRBs for these UEs = $\frac{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3} + \text{L.ChMeas.MIMO.PRB.CL.Rank1} + \text{L.ChMeas.MIMO.PRB.CL.Rank2} + \text{L.ChMeas.MIMO.PRB.CL.Rank3})}{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3} + \text{L.ChMeas.BF.PRB.OL.Rank4} + \text{L.ChMeas.MIMO.PRB.CL.Rank1} + \text{L.ChMeas.MIMO.PRB.CL.Rank2} + \text{L.ChMeas.MIMO.PRB.OL.Rank1} + \text{L.ChMeas.MIMO.PRB.OL.Rank2} + \text{L.ChMeas.MIMO.PRB.CL.Rank3} + \text{L.ChMeas.MIMO.PRB.OL.Rank3} + \text{L.ChMeas.MIMO.PRB.OL.Rank4} + \text{L.ChMeas.MIMO.PRB.CL.Rank5} + \text{L.ChMeas.MIMO.PRB.CL.Rank6} + \text{L.ChMeas.MIMO.PRB.CL.Rank7} + \text{L.ChMeas.MIMO.PRB.CL.Rank8})}$

Beamforming UEs using open-loop TM7, TM8, or TM9 (excluding UEs using closed-loop TM9) Such UEs do not include UEs with rank 4 or above.

Proportion of PRBs for these UEs = $\frac{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3})}{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3} + \text{L.ChMeas.BF.PRB.OL.Rank4} + \text{L.ChMeas.MIMO.PRB.CL.Rank1} + \text{L.ChMeas.MIMO.PRB.CL.Rank2} + \text{L.ChMeas.MIMO.PRB.OL.Rank1} + \text{L.ChMeas.MIMO.PRB.OL.Rank2} + \text{L.ChMeas.MIMO.PRB.CL.Rank3} + \text{L.ChMeas.MIMO.PRB.CL.Rank4} + \text{L.ChMeas.MIMO.PRB.OL.Rank3} + \text{L.ChMeas.MIMO.PRB.OL.Rank4})}$
- The last TTI before the downlink buffer is emptied occupies less than 40% of the data transmission duration.

Proportion of the last TTI before the downlink buffer is empty in LTE cells =
 $(L.Thrp.Time.DL - L.Thrp.Time.DL.RmvLastTTI)/L.Thrp.Time.DL$

Proportion of the last TTI before the downlink buffer is empty in NR cells =
 $(N.ThpTime.DL - N.ThpTime.DL.RmvLastSlot)/N.ThpTime.DL$

NOTE

- In certain scenarios where NR does not support power sharing-out, only LTE-to-NR unidirectional power sharing is available. In such scenarios, if the power sharing group includes only one LTE cell, the LTE cell does not have any gains; if the power sharing group includes multiple LTE cells, the gains of LTE cells are decreased. For details about NR not supporting power sharing-out, see [Determining the Power Sharing Status](#).
- A lower load of the shared-out cell results in a higher probability of power sharing-out and a larger amount of shared-out power, as well as a higher probability of power sharing-in for shared-in cells and hence larger power-sharing benefits. A higher load of the shared-out cell results in the opposite effects.
- When the configured power of shared-in cells is fixed, a larger configured power of shared-out cells indicates a larger amount of shared-out power, as well as a larger amount of shared-in power for shared-in cells and hence larger power-sharing benefits. A smaller configured power of shared-out cells results in the opposite effects.
- If the amount of shared-in power is fixed, a smaller configured power of shared-in cells indicates a larger power sharing gain. A larger configured power of shared-in cells results in the opposite effects.
- If there are multiple beneficiary cells (for example, both NR and LTE cells) and one of the cells (for example, an NR cell) is configured with large power or is heavily loaded, the cell preferentially uses the shared power. In this case, this cell obtains larger gains than other cells, where even no gains are achieved.

6.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has the following impacts on LTE and NR network performance:

- This function increases the power consumed by the PDSCH, possibly increases the average MCS index, and reduces the downlink PRB usage.
- The impact of this function on the downlink packet delay (LTE: **L.Traffic.DL.PktDelay.Time**; NR: **N.QoS.DL.PktDelaygNBDU.Time**) is related to services. When the tail packet percentage (the proportion of the last TTI before the downlink buffer is emptied) is low, the downlink packet delay decreases. When the tail packet percentage is high, the downlink packet delay increases.
- When this function is enabled, the average uplink and downlink UE throughput may decrease for cells that constantly share out power.
 - LTE: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput**.
 - NR: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput (DU)**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput (DU)**.
- Function activation or deactivation involves the adjustment of system parameters, which causes service rate fluctuations for approximately 2

minutes and affects the cell throughput. Therefore, it is recommended that you enable this function when the traffic is light.

- When this function is enabled, the RF module power utilization increases. Base stations consume more power, but the total power consumption does not exceed the maximum power consumed when all cells work with a full load.
- This function requires information exchanges between cells. This may increase the CPU load of baseband processing units (**VS.BBUBoard.CPULoad.Mean**) by less than 5%.
- This function increases the instantaneous power used for PDSCH, which may increase interference to neighboring cells. It is recommended that this function be enabled in contiguous coverage areas. If the average cell throughput decreases, or if the interference to neighboring cells significantly increases after this function is enabled, you can decrease the maximum power sharing rate to counteract the negative impacts. (Setting the maximum power sharing rate to half of its recommended value is recommended.) If the gains are not noticeable, you can increase the value of the maximum power sharing rate to achieve higher function gains.
 - LTE: The maximum power sharing rate is specified by the **CellDynPowerSharing.MaxPowerSharingRate** parameter.
 - NR: The maximum power sharing rate is specified by the **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** parameter.
- It is recommended that this function be disabled before downlink peak rate tests on an eNodeB or gNodeB.
- When the **LNR_POWER_SHARING_RATIO_OPT_SW** option of the **CellDynPowerSharing.DynamicPowerSharingSwitch** parameter is selected, as scheduling is performed in advance on the LTE side, the random access response success rate slightly decreases, and the UE access delay slightly increases. The average uplink and downlink UE throughputs slightly decrease as the LTE cell has its power shared.

6.3 Requirements

6.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE TDD	TDLCOFD-221611	L/NR Inter-band Deep Coordination (LTE TDD)	LT4SNTBADC00	per Cell
NR TDD	FOFD-101225	L/NR Inter-band Deep Coordination	NR0SNTBACD00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

The prerequisite functions on the LTE side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	L/NR Inter-band Deep Coordination requires LTE dynamic power sharing to be enabled. LTE cells enabled with Dynamic Power Sharing Between LTE Carriers automatically form an LTE power sharing group when the conditions for triggering Dynamic Power Sharing Between LTE Carriers are met. When the LTE cells included in the L/NR Inter-band Deep Coordination group are the same as those included in the LTE power sharing group, both of these two functions take effect. Otherwise, only Dynamic Power Sharing Between LTE Carriers takes effect, and L/NR Inter-band Deep Coordination does not take effect. For details about Dynamic Power Sharing Between LTE Carriers, see <i>Dynamic Power</i>

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
				<i>Sharing Between LTE Carriers.</i>

The prerequisite functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgorithmSwitch.DynamicPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	L/NR Inter-band Deep Coordination requires NR inter-carrier dynamic power sharing to be enabled. Cells enabled with NR inter-carrier dynamic power sharing automatically form an NR power sharing group when the conditions for triggering NR inter-carrier dynamic power sharing are met. When the LTE cells included in the L/NR Inter-band Deep Coordination group are the same as those included in the NR power sharing group, both of these two functions take effect. Otherwise, only NR inter-carrier dynamic power sharing takes effect, and L/NR Inter-band Deep Coordination does not take effect.

6.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions on the LTE side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	L/NR Inter-band Deep Coordination cannot be enabled together with LTE TDD and NR Flash Dynamic Power Sharing.
LTE TDD	PHICH duration	PHICHCfg.PhiChDuration set to EXTENDED	<i>Physical Channel Resource Management</i>	L/NR Inter-band Deep Coordination cannot be enabled together with PHICH duration.
LTE TDD	PDCCH spare power allocation optimization	CellPdcchAlgorithm.PdcchSparePowerAllocation	<i>Physical Channel Resource Management</i>	L/NR Inter-band Deep Coordination cannot be enabled together with PDCCH spare power allocation optimization.
LTE TDD	PDCCH power check	CellPdcchAlgorithm.PdcchPowerCheckSwitch	None	L/NR Inter-band Deep Coordination cannot be enabled together with PDCCH power check.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Soft split resource duplex	None	<i>Soft Split Resource Duplex (TDD)</i>	If Cell.MultiCellShareMode is set to SPLIT for a cell and the SsrdCellGroup MO with SsrdCellGroup.PrimaryLocalCellID or SsrdCellGroup.SsrdLocalCellID set to the value of Cell.LocalCellId of the cell, L/NR Inter-band Deep Coordination must be disabled.
LTE TDD	CSPC	CellCspcPara.CellCspcSwitch	<i>CSPC</i>	L/NR Inter-band Deep Coordination cannot be enabled together with CSPC.
LTE TDD	Enhanced coordinated scheduling based power control	CellCspcPara.CelleCspcSwitch	<i>Massive MIMO Enhancements (TDD)</i>	L/NR Inter-band Deep Coordination cannot be enabled together with enhanced coordinated scheduling based power control.
LTE TDD	PHICH inner-loop power control	PhichInnerLoopPcSwitch option of the CellAlgoSwitch.DIPcAlgoSwitch parameter	<i>Power Control</i>	L/NR Inter-band Deep Coordination cannot be enabled together with PHICH inner-loop power control.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	PDCCH SDMA enhancement	CellPdcchAlgo.PdcchSdmaWeightType set to POWER_SHARING_AMONG_3_BEAMS , POWER_SHARING_AMONG_4_BEAMS , or POWER_SHARING_AMONG_5_BEAMS	<i>Massive MIMO Basics (TDD)</i>	If PDCCH SDMA (controlled by the MASSIVE_MIMO_PDCCH_SDMA_SW option of CellPdcchAlgo.SplitBeamPdcchSdmaSw) and L/NR Inter-band Deep Coordination are both enabled, PDCCH SDMA enhancement cannot be enabled.
LTE TDD	B-Trunc mode	eNodeBFunction.eNodeBMode set to BTRUNC_MODE	None	L/NR Inter-band Deep Coordination cannot be enabled together with B-Trunc mode.
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD)</i>	If Cell.TxRxMode is set to 8T8R and a SectorSplitCell or SectorSplitCellExt MO has been configured, L/NR Inter-band Deep Coordination cannot be enabled.
LTE TDD	Flag of Multi-RRU Cell	Cell.MultiRruCellFlag set to BOOLEAN_TRUE	<i>SFN</i>	If Cell.TxRxMode is set to 8T8R and Cell.MultiRruCellFlag is set to BOOLEAN_TRUE , L/NR Inter-band Deep Coordination cannot be enabled.

The mutually exclusive functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	If L/NR Inter-band Deep Coordination is enabled together with High-speed Railway Superior Experience, power estimation may be inaccurate. Therefore, they cannot be both enabled.
Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	L/NR Inter-band Deep Coordination cannot be enabled for hyper cells.
Low-frequency NR TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	L/NR Inter-band Deep Coordination cannot be enabled together with Fusion Cell.
Low-frequency NR TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	L/NR Inter-band Deep Coordination cannot be enabled together with Bi-Carrier.
Low-frequency NR TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgorithmSwitch.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	L/NR Inter-band Deep Coordination cannot be enabled together with LTE TDD and NR Flash Dynamic Power Sharing.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	NR smart power lock	AVERAGE_POWER_CONTROL_SW option of the NRDUCellSmartPwrLock.SmartPowerLockSw parameter	None	If enhanced NR power sharing is enabled by selecting the NR_PWR_SHARE_CAP_ENH_SW option of NRDUCellResShrCfg.DynPwrShrAlgoSwitch , the power sharing capability exceeds 100%. In this case, the power is more likely to be limited when smart power lock is also enabled, affecting cell performance. Therefore, these two functions cannot be both enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	NR smart power scale	AVERAGE_3D EIRP_CONTR OL_SW option of the NRDUCellSmartPwrLock.S martPowerLo ckSw parameter	None	If enhanced NR power sharing is enabled by selecting the NR_PWR_SHARE_CAP_ENH_SW option of NRDUCellResShr Cfg.DynPwrShrAl goSwitch , the power sharing capability exceeds 100%. In this case, the power is more likely to be limited when smart power lock is also enabled, affecting cell performance. Therefore, these two functions cannot be both enabled.

6.3.2.3 Function Impacts

The function impacts on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LTE TDD	Scalable bandwidth	None	<i>Cell Management</i>	10 MHz LTE cells that use default channel configurations seldom share power with NR cells as these LTE cells have only few power resources left after power reservation for paging channels, control channels, and traffic bursts.
LTE TDD	DL 256QAM	CellAlgoSwitch.Dl256QamAlgoSwitch	<i>Modulation Schemes</i>	When UEs for which DL 256QAM has taken effect are scheduled in a TTI on the LTE side, LTE does not use the power shared by NR.
LTE TDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDy nMuting.RfChnDynMutin gAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	L/NR Inter-band Deep Coordination does not take effect when RF channel dynamic muting is in effect.
LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_S CH_ENERGY_SAVING_SW option of the CellPwrSavin gAlgo.PwrSavin gAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Energy saving based on flexible frequency-domain scheduling is not performed in subframes where received shared power is used.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown . CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown . CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Multi-carrier coordinated energy saving	CellShutdown . CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIER_MANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynmicPowerSharingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynmicPowerSharingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPTION_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPTION_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSav gAlgo.Carrier DynShutdow nPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.Dyna micPowerShar ingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.Dyna micPowerShar ingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Cells that have entered the energy saving state cannot participate in L/NR Inter-band Deep Coordination. These cells will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes.
LTE TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over L/NR Inter-band Deep Coordination when both functions are enabled. L/NR Inter-band Deep Coordination does not take effect when RF module fast deep dormancy is in effect.

The function impacts on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	When UEs for which DL 256QAM has taken effect are scheduled in a slot on the NR side, NR does not use the power shared by LTE.
Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When L/NR Inter-band Deep Coordination and power saving scheduling are both enabled, the effects of power saving scheduling are affected. It is recommended that L/NR Inter-band Deep Coordination not be enabled with this function when the traffic of LTE and NR cells is light.
Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When L/NR Inter-band Deep Coordination takes effect together with energy saving based on flexible frequency-domain scheduling, the benefits of both functions decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Multi-position PDCCH scheduling	ANY_OCCASION_PDCCH_Switch.LowLatencySwitch parameter	None	When L/NR Inter-band Deep Coordination and multi-position PDCCH scheduling are both enabled, NR cells cannot estimate the slot and power resources to be scheduled for UEs for which multi-position PDCCH scheduling takes effect. Consequently, NR cells can receive shared power, and do not share power to other cells.
Low-frequency NR TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_Switch option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When L/NR Inter-band Deep Coordination and intra-base-station DL CoMP are both enabled, the power sharing gains on the NR side may decrease.
Low-frequency NR TDD	Co-MM	INTRA_GNB_SU_COMM_Switch option of the NRDUCellColLabServ.MultiCellMIMOSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When L/NR Inter-band Deep Coordination and Co-MM are both enabled, the power sharing gains on the NR side may decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	When L/NR Inter-band Deep Coordination and coordinated interference management are both enabled, the power sharing gains on the NR side may decrease.
Low-frequency NR TDD	Power reduction for CCUs	None	<i>Power Control</i>	When L/NR Inter-band Deep Coordination and power reduction for CCUs are both enabled, the power sharing gains on the NR side may decrease.
Low-frequency NR TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW or INTRF_COORD_ALWAYS_WORKING_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	When L/NR Inter-band Deep Coordination and "interference coordination between public and dedicated networks" are both enabled, the power sharing gains on the NR side may decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of timing carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of intelligent carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH,

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter is deselected for a cell, power sharing is not performed on the cell within the period specified for RF channel intelligent shutdown. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter is selected for a cell, power sharing is not performed on the cell when the cell is in the energy saving state. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Fast carrier shutdown	- Capacity-layer cells: - In intra-base-station multi-carrier scenario s: INTRA_GNB_MULTICARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter and FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter - In inter-base-station multi-carrier scenario	<i>Energy Conservation and Emission Reduction</i>	- If the PWR_SHR_IN_PWR_SAVE_SWITCH option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of fast carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. - If the PWR_SHR_IN_PWR_SAVE_SWITCH option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
		S: INTER_GNB_MULTICAST_ARR_SD_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingSwitch</i> parameter and FAST_CARRIER_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter		messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	TTI-level channel shutdown	NRDUCellPowerSaving.CombChnDynShutdownSwitch parameter and TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When L/NR Inter-band Deep Coordination takes effect together with TTI-level channel shutdown, the energy saving benefits of TTI-level channel shutdown decrease and downlink UE throughput may decrease.
Low-frequency NR TDD	PDCCH aggregation level reduction	PDCCH_AGG_LVL_COMPRESS_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When L/NR Inter-band Deep Coordination and PDCCH aggregation level reduction are both enabled, only L/NR Inter-band Deep Coordination takes effect.
Low-frequency NR TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellIDMAC.DLModulationSchemeSwitch parameter	<i>Modulation Schemes</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where L/NR Inter-band Deep Coordination and LTE and NR power sharing optimization take effect decreases.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Only low power consumption mode takes effect when low power consumption mode and L/NR Inter-band Deep Coordination are both enabled and a customized backup power saving policy is used in low power consumption mode.
Low-frequency NR TDD	Maximum SSB power offset	NRDUCellTrpBeam.MaxSsbPwrOffset	<i>Power Control</i>	When the SSB power offset is set to a value greater than 3 dB and L/NR Inter-band Deep Coordination is in effect, the SSB power aggregation capability decreases.
Low-frequency NR TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When L/NR Inter-band Deep Coordination takes effect, the benefits of CCE partial symbol adaptation will decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of RF module fast deep dormancy arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

LTE: 3900 and 5900 series base stations (macro base stations)

NR: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

- Main control boards: All main control boards that support 8T8R and eCPRI-capable main control boards that support 32T32R or higher support this function. For details, see section "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.
- Baseband processing units: UBBPe and later baseband processing units

RF Modules

RF modules that meet both of the following conditions support this function:

- The RF modules are 8T8R RRUs or eCPRI-capable 32T32R or higher AAUs.
- The RF modules must support LTE and NR concurrency.

6.3.4 Networking

This function has the following networking requirements:

- If an L/NR Inter-band Deep Coordination group consists of both LTE and NR cells, the LTE and NR cells must be served by the same group of RF channels, and the LTE subframe configuration, NR slot assignment, and NR slot structure must meet the requirements listed in [Table 6-3](#).

Table 6-3 Requirements for the LTE subframe configuration, NR slot assignment, and NR slot structure

No.	LTE Subframe Configuration	LTE Special Subframe Configuration	NR Slot Assignment	NR Slot Structure	Base Station Model
1	SA2	SSP7	Single-period: 8:2 Dual-period: 8:2	Single-period: SS54 Dual-period: SS84	Macro base station
2	SA2	SSP6	Single-period: 8:2 Dual-period: 8:2	Single-period: SS56 Dual-period: SS86	Macro base station
3	SA2	SSP5	Single-period: 8:2 Dual-period: 8:2	Single-period: SS518 Dual-period: SS818	Macro base station

For single period configuration, the NR cell has a 3 ms (92160 Ts) frame offset relative to the LTE cell. For dual-period configuration, the NR cell frame offset is aligned with the LTE cell frame offset. For details about the NR slot assignment and structure, see *Standards Compliance* in *5G RAN Feature Documentation*.

- LTE and NR cells support only 8T8R, 32T32R, and 64T64R.
- The LTE cells for which L/NR Inter-band Deep Coordination is enabled must be the same as those for which Dynamic Power Sharing Between LTE Carriers is enabled; otherwise, L/NR Inter-band Deep Coordination cannot take effect. The NR cells for which L/NR Inter-band Deep Coordination is enabled must be the same as those for which NR inter-carrier dynamic power sharing is enabled; otherwise, L/NR Inter-band Deep Coordination cannot take effect.

- When multiple NR carriers are configured on an RF channel, L/NR Inter-band Deep Coordination can be enabled only on two NR carriers.

6.3.5 Cells

The requirements for LTE cells are listed below.

- Cells cannot form an L/NR Inter-band Deep Coordination group if there are carriers of less than 5 MHz on the RF channels.
- L/NR Inter-band Deep Coordination can be enabled only when the cell bandwidth is greater than or equal to 5 MHz.
- The eMTC-only cell cannot be added to an L/NR Inter-band Deep Coordination group to perform power sharing.
- The **Cell.FddTddInd** parameter is set to **CELL_TDD**.
- The **Cell.TxRxMode** parameter is set to **64T64R**, **32T32R**, or **8T8R**.

The requirements for NR cells are listed below.

- The **NRDUCell.DuplexMode** parameter is set to **CELL_TDD**.
- The **NRDUCell.SubcarrierSpacing** parameter is set to **30KHZ**.
- The **NRDUCellTrp.TxRxMode** parameter is set to **64T64R**, **32T32R**, or **8T8R**.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

This feature takes effect only after parameters are configured on both the eNodeB and gNodeB sides. [Table 6-4](#) and [Table 6-5](#) describe the parameters used for function activation on the eNodeB and gNodeB sides, respectively.

Table 6-4 eNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	CellDynPowerSharing.DynamicPowerSharingSwitch	• Select the LNR_INTER_BAND_DEEP_COORD_SW option. • Select the LNR_POWER_SHARING_RATIO_OPT_SW option.
Max Power Sharing Rate	CellDynPowerSharing.MaxPowerSharingRate	The default value is recommended.
Multi-Band Power Adapt Scheduling Priority	CellDynPowerSharing.MultiBandPwrAdaptSchPri	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Power Assistance Switch	CellChPwrCfg.PwrAssistanceSwitch	It is recommended that the PWR_SHR_ASSIST_SW option of this parameter be selected to increase the maximum power sharing percentage when L/NR Inter-band Deep Coordination is enabled.
Interference Control Switch	CellDynPowerSharing.InterferenceControlSwitch	It is recommended that this parameter be set to ON for the local cell and its intra-frequency neighboring cells when L/NR Inter-band Deep Coordination is enabled.

Table 6-5 gNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Algorithm Switch	NRDUCellResShrCfg.DynPwrShrAlgorithm	Select the LNR_INTER_BAND_DEEP_COORD_SW option.
Dynamic Power Sharing Algorithm Switch	NRDUCellResShrCfg.DynPwrShrAlgorithm	Select the NR_PWR_SHARE_CAP_ENH_SW option if an AAU5685md is used.
Max Power Sharing Rate	NRDUCellTrpDynPwrShr.MaxPowerSharingRate	The default value is recommended.
Multi-Band Power Adapt Scheduling Priority	NRDUCellTrpDynPwrShr.MultiBandPwrAdaptSchPri	Use the recommended value.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//On the eNodeB side
MOD CELLDYNPOWERSHARING: LocalCellId=0,
DynamicPowerSharingSwitch=LNR_INTER_BAND_DEEP_COORD_SW-1, MaxPowerSharingRate=50,
MultiBandPwrAdaptSchPri=0;
MOD CELLDYNPOWERSHARING: LocalCellId=0,
DynamicPowerSharingSwitch=LNR_POWER_SHARING_RATIO_OPT_SW-1;
MOD CELLCHPWRCFG: LocalCellId=0, PwrAssistanceSwitch=PWR_SHR_ASSIST_SW-1;
MOD CELLDYNPOWERSHARING: LocalCellId=0, InterferenceControlSwitch=ON;

//On the gNodeB side
MOD NRDUCELLRESSHRCFG: NrDuCellId=0, DynPwrShrAlgoSwitch=LNR_INTER_BAND_DEEP_COORD_SW-1;
MOD NRDUCELLTRPDYNPWRSHR: NrDuCellTrpld=0, MaxPowerSharingRate=50,
MultiBandPwrAdaptSchPri=0;
//When an AAU5685md is used
MOD NRDUCELLRESSHRCFG: NrDuCellId=0, DynPwrShrAlgoSwitch=NR_PWR_SHARE_CAP_ENH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//On the eNodeB side
MOD CELLDYNPOWERSHARING: LocalCellId=0, InterferenceControlSwitch=OFF;
MOD CELLCHPWRCFG: LocalCellId=0, PwrAssistanceSwitch=PWR_SHR_ASSIST_SW-0;
MOD CELLDYNPOWERSHARING: LocalCellId=0,
DynamicPowerSharingSwitch=LNR_POWER_SHARING_RATIO_OPT_SW-0;
MOD CELLDYNPOWERSHARING:
LocalCellId=0,DynamicPowerSharingSwitch=LNR_INTER_BAND_DEEP_COORD_SW-0;

//On the gNodeB side
MOD NRDUCELLRESSHRCFG: NrDuCellId=0, DynPwrShrAlgoSwitch=NR_PWR_SHARE_CAP_ENH_SW-0;
MOD NRDUCELLRESSHRCFG: NrDuCellId=0, DynPwrShrAlgoSwitch=LNR_INTER_BAND_DEEP_COORD_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Perform the following steps to check whether this function has taken effect.

1. Check whether this function has taken effect in LTE cells.
Run the eNodeB MML command **DSP CELLDYNPWRSHASTATUS**. If the value of **LNR Inter-Band Deep Coord Status** is **EFFECTIVE**, this function has taken effect in LTE cells.
2. Check whether this function has taken effect in NR cells.
Run the gNodeB MML command **DSP NRDUCELLTRPDYNPWRSHR**. If the value of **LNR Inter-band Deep Coord Valid Status** is **Enabled**, this function has taken effect in NR cells.
3. Monitor the values of the counters listed in the tables below on the MAE. If the counter values are greater than 0, this function has taken effect. If the counter values are 0, this function has not taken effect.

Table 6-6 eNodeB counters

Counter ID	Counter Name
1526746987	L.PwrShare.PwrOut.TTI.Num
1526746986	L.PwrShare.PwrIn.TTI.Num

Table 6-7 gNodeB counters

Counter ID	Counter Name
1911823028	N.PwrShare.PwrOut.Slot.Num
1911823027	N.PwrShare.PwrIn.Slot.Num

 NOTE

- After L/NR Inter-band Deep Coordination is activated, LTE and NR cells will negotiate the configuration of dynamic power sharing. Generally, the negotiation can be completed within 5 minutes after the last activation step. Therefore, it is recommended that the activation status be queried 5 minutes after the activation procedure is complete.
- Intra-RAT power sharing is preferentially performed in an L/NR Inter-band Deep Coordination group. If there is still surplus power after intra-RAT power sharing, inter-RAT power sharing can be performed. The performance counters related to power sharing measure the actual power sharing-in or power sharing-out time of each cell, without distinguishing between intra-RAT and inter-RAT power sharing.
- LTE measures the number of TTIs in which a cell shares out power and the number of TTIs in which a cell receives shared power in each 1-ms TTI. NR measures the number of slots in which a cell shares out power and the number of slots in which a cell receives shared power.
- The **L.PwrShare.PwrIn.TTI.Num** counter indicates the number of TTIs within which the cell uses shared power. As power sharing uses the estimation results from the previous TTI, the estimated number of TTIs within which the cell has received shared power might be incorrect.
- The **N.PwrShare.PwrIn.Slot.Num** counter indicates the number of slots within which the cell uses shared-in power. As power sharing uses the estimation results from the previous millisecond, the estimated number of slots within which the cell has received shared power might be incorrect.
- When a slot of an NR cell is counted as a power sharing-in slot, the actual power used in the slot is generally greater than the statically configured cell power. However, there are also exceptions. When PDSCH power aggregation is disabled for an NR cell, the actual power used in the slot is less than the statically configured cell power if there are remaining RBs in the slot and the power of the remaining RBs is greater than the increased power of beneficiary UEs.
- When the power of a single cell is shared with one or multiple cells in an LTE and NR power sharing group, the number of TTIs in which a cell uses shared power is counted for each cell. Therefore, the number of TTIs in which a cell shares its power with other cells differs from the number of TTIs in which a cell uses shared-in power.

6.4.3 Network Monitoring

LTE Network Monitoring

When event A3 can be reported, determine whether this feature has taken effect based on the throughput of CEUs, which can be calculated based on the information contained in event A3 reports. The detailed steps are as follows:

Step 1 One week before feature activation, select the **BasedA3EdgeUserSwitch** option of the **CellCounterParaGroup.CellCounterAlgoSwitch** parameter, and set the **CellCounterParaGroup.EdgeUserA3Offset** parameter to **-12**.

 NOTE

The average downlink throughput of CEUs can be measured only when the **BasedA3EdgeUserSwitch** option of the **CellCounterParaGroup.CellCounterAlgoSwitch** parameter is selected.

Step 2 Observe the values of the following counters to monitor feature gains.

- Average downlink UE throughput = $(L.Thrp.bits.DL - L.Thrp.bits.DL.LastTTI) / L.Thrp.Time.DL.RmvLastTTI$
- Average downlink throughput of CEUs = $(L.Thrp.bits.DL.BorderUE - L.Thrp.bits.DL.LastTTI.BorderUE) / L.Thrp.Time.DL.RmvLastTTI.BorderUE$

----End

When event A3 cannot be reported, determine whether this feature has taken effect based on the cumulative distribution function (CDF) curve of the throughput.

Create the CDF curve based on values of counters listed in [Table 6-8](#). If the throughput corresponding value 5% or 10% on the CDF curve increases, this feature has taken effect.

Table 6-8 LTE counters

Counter ID	Counter Name
1526729056 to 1526729065	L.Thrp.DL.BitRate.Samp.Index0 to L.Thrp.DL.BitRate.Samp.Index9

NR Network Monitoring

Step 1 Select the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter one week before feature activation.

 NOTE

The average downlink throughput of UEs in overlapping areas can be measured only when the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter is selected.

Step 2 Observe the values of the following counters to monitor feature gains.

- Average downlink UE throughput = $(N.ThpVol.DL - N.ThpVol.DL.LastSlot) / N.ThpTime.DL.RmvLastSlot$
- Average downlink throughput of UEs in overlapping areas = $(N.ThpVol.DL.CellOverlap - N.ThpVol.DL.LastSlot.CellOverlap) / N.ThpTime.DL.RmvLastSlot.CellOverlap$

----End

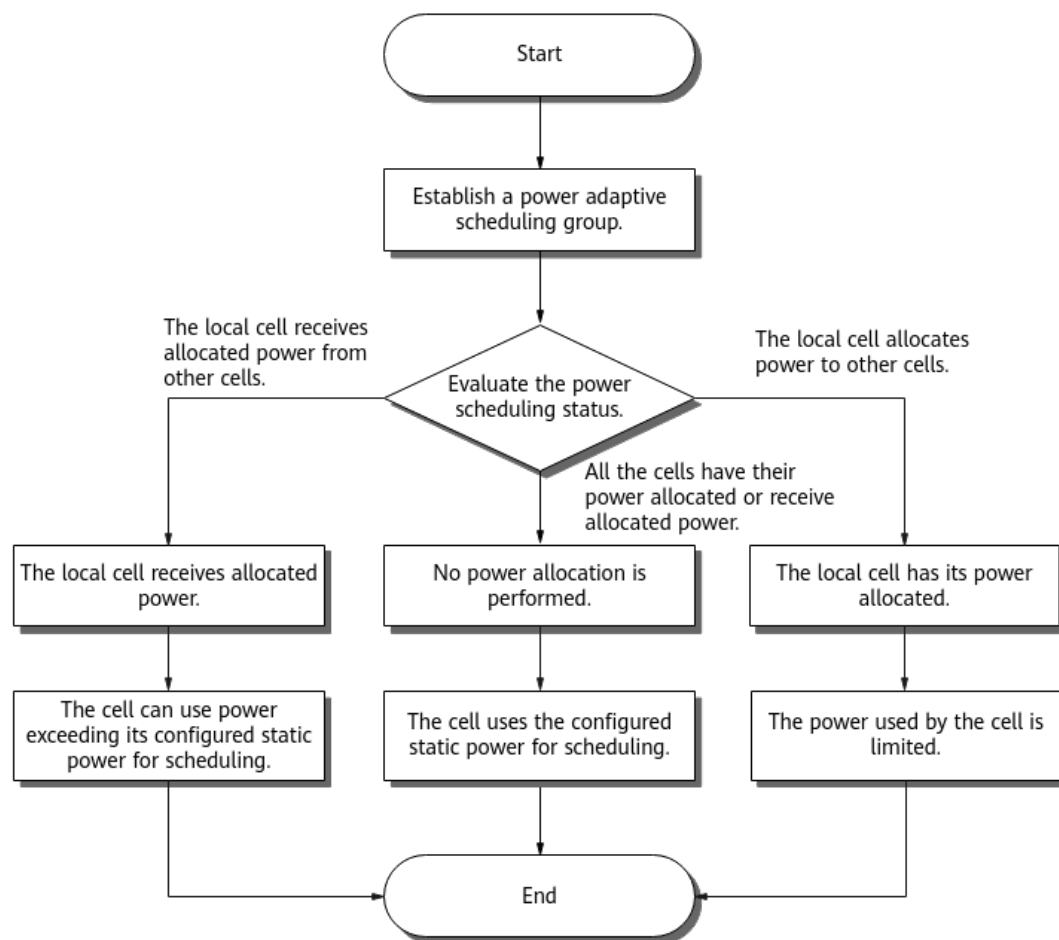
7

LTE and NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)

7.1 Principles

LTE and NR multi-band power adaptive scheduling enables TTI-level power adaptive scheduling within an RF module. [Figure 7-1](#) shows the process.

Figure 7-1 Process of LTE and NR multi-band power adaptive scheduling



This function requires the following options on both the LTE and NR sides to be selected:

- LTE: **LNR_MULTI_BAND_PWR_ADPT_SCH_SW** option of the **CellDynPowerSharing.DynamicPowerSharingSwitch** parameter
- NR: **LNR_MULTI_BAND_PWR_ADPT_SCH_SW** option of the **NRDUCellAlgoSwitch.DynPowerSharingSwitch** parameter

Process of LTE and NR Multi-Band Power Adaptive Scheduling (TTI-Level)

1. Establishment of power adaptive scheduling groups

After this function is enabled, the gNodeB and eNodeB automatically set up power adaptive scheduling groups for NR and LTE cells set up on RF modules when all of the following conditions are met:

- RF modules meet the requirements described in [RF Modules](#).
- In a single frequency band, the cells have the same number of downlink transmit channels.
- In a single frequency band, the cells share power amplifiers (PAs).

Each baseband processing unit supports a maximum of six LTE and NR multi-band power adaptive scheduling groups. Each power sharing group supports power sharing between a maximum of five LTE cells and four NR cells. LTE

cells must work on the same frequency band. Otherwise, this function cannot be enabled. The AAU5368md is used as an example. It can work on the 2.3 GHz and 3.5 GHz frequency bands. All LTE cells served by this module must work on the 2.3 GHz or 3.5 GHz frequency band so that this function can be enabled.

2. Evaluation of the power adaptive scheduling status of each cell

The eNodeB and gNodeB check the status of each cell in the power adaptive scheduling group based on the to-be-scheduled data volume estimated in advance and evaluate whether a cell receives allocated power, allocates its power, or does not participate in power adaptive scheduling.

3. Calculation of the power used by a cell

LTE and NR multi-band power adaptive scheduling cannot increase the RF channel capability, but it can increase user-perceived throughput by reallocating power from light-load cells to heavy-load cells. When this function is enabled, the sum of the power used by all of the cells in a power adaptive scheduling group cannot exceed the sum of their configured static power or the maximum rated power of the RF module channels.

- For a cell receiving allocated power:

TTI-level power used by a cell = Min{Configured static power + Received allocated power, Maximum rated power of RF module channels}

TTI-level maximum allocated power received by a cell = Max{Configured static power x Maximum percentage of allocated power, Maximum rated power of RF module channels}

The maximum percentage of allocated power is specified by the following parameters:

- LTE: The maximum percentage of allocated power is specified by the **CellDynPowerSharing.MaxPowerSharingRate** parameter.
- NR: The maximum percentage of allocated power is specified by the **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** parameter.

The priority of a carrier to receive allocated power in a power adaptive scheduling group is specified by the following parameters. In such a group, a smaller parameter value indicates a lower priority and a larger value indicates a higher priority.

- LTE: **CellDynPowerSharing.MultiBandPwrAdaptSchPri**
- NR: **NRDUCellTrpDynPwrShr.MultiBandPwrAdaptSchPri**

NOTE

"Configured static power" refers to the power manually configured for a cell.

"Maximum rated power of RF module channels" refers to the maximum rated power of the frequency band of the local cell served by the RF module channels.

- For a cell having its power allocated: Power used by the cell in a TTI = Configured static power – Power allocated
- For a cell not participating in power scheduling: Power used by the cell in a TTI = Configured static power

7.2 Network Analysis

7.2.1 Benefits

It is recommended that this function be enabled when an RF module serves multiple LTE and NR cells that do not share channels and work in different frequency bands, and traffic is instantaneously imbalanced between these cells. The specific benefits provided by this function are as follows:

- NR: The **User Downlink Average Throughput (DU)** increases by 2–10%.
- LTE: The **User Downlink Average Throughput** increases by 1–5%.

To achieve optimal benefits, all of the following conditions must be met:

- The last TTIs before the downlink buffer is empty occupy less than 40% of the data transmission duration.

Proportion of the last TTIs before the downlink buffer is empty in LTE cells = $(L.Thrp.Time.DL - L.Thrp.Time.DL.RmvLastTTI) / L.Thrp.Time.DL$

Proportion of the last TTIs before the downlink buffer is empty in NR cells = $(N.ThpTime.DL - N.ThpTime.DL.RmvLastSlot) / N.ThpTime.DL$

- [Table 7-1](#) lists the requirements for the downlink PRB usage on the LTE side and that on the NR side.

Table 7-1 Requirements for PRB usage in different scenarios

Combination of LTE and NR Cells	Beneficiary Scenario	Requirement for LTE Downlink PRB Usage	Requirement for NR Downlink PRB Usage
One NR cell + one LTE cell	Benefits achieved on the LTE side	LTE downlink PRB usage > 50%	NR downlink PRB usage < 30%
	Benefits achieved on the NR side	LTE downlink PRB usage < 30%	NR downlink PRB usage > 50%
One NR cell + multiple LTE cells	Benefits achieved on the LTE side	LTE downlink PRB usage < 40%	NR downlink PRB usage < 30%
	Benefits achieved on the NR side	LTE downlink PRB usage < 30%	NR downlink PRB usage > 50%
Multiple NR cells + one LTE cell	Benefits achieved on the LTE side	LTE downlink PRB usage > 50%	NR downlink PRB usage < 30%
	Benefits achieved on the NR side	LTE downlink PRB usage < 40%	NR downlink PRB usage < 40%

Combination of LTE and NR Cells	Beneficiary Scenario	Requirement for LTE Downlink PRB Usage	Requirement for NR Downlink PRB Usage
Multiple NR cells + multiple LTE cells	Benefits achieved on the LTE side	LTE downlink PRB usage < 40%	NR downlink PRB usage < 30%
	Benefits achieved on the NR side	LTE downlink PRB usage < 40%	NR downlink PRB usage < 40%

For example, in scenarios where there is one NR cell and one LTE cell, to achieve benefits on the LTE side, the LTE downlink PRB usage must be greater than 50% and the NR downlink PRB usage must be less than 30%.

Average downlink PRB usage on the LTE side = $\frac{\text{L.ChMeas.PRB.DL.Used.Avg}}{\text{L.ChMeas.PRB.DL.Avail}}$

Average downlink PRB usage on the NR side = $\frac{\text{N.PRB.DL.Used.Avg}}{\text{N.PRB.DL.Avail.Avg}}$

- Any of the following groups of UEs accounts for at least 20% of the UEs in LTE cells:
 - UEs using QPSK modulation without involving signaling transmission and MCS selection with prioritized RBs
Proportion of UEs for which QPSK is used = $\frac{\text{Sum of the values of counters L.ChMeas.PDSCH.MCS.0 to L.ChMeas.PDSCH.MCS.9}}{\text{Sum of the values of counters L.ChMeas.PDSCH.MCS.0 to L.ChMeas.PDSCH.MCS.28}}$

NOTE

If the *CellDischAlgo.RBPrMcsSelectRatioThd* parameter is not set to 0, the proportion of UEs for which QPSK is used approximates to the sum of the values of counters *L.ChMeas.PDSCH.MCS.2* to *L.ChMeas.PDSCH.MCS.9* divided by the sum of the values of counters *L.ChMeas.PDSCH.MCS.0* to *L.ChMeas.PDSCH.MCS.28*.

- Beamforming UEs using open-loop TM7, TM8, or TM9 and UEs using closed-loop TM9. Open-loop TM9 UEs do not include UEs with rank 4 or above.
Proportion of PRBs for these UEs = $\frac{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3} + \text{L.ChMeas.MIMO.PRB.CL.Rank1} + \text{L.ChMeas.MIMO.PRB.CL.Rank2} + \text{L.ChMeas.MIMO.PRB.CL.Rank3})}{(\text{L.ChMeas.BF.PRB.OL.Rank1} + \text{L.ChMeas.BF.PRB.OL.Rank2} + \text{L.ChMeas.BF.PRB.OL.Rank3} + \text{L.ChMeas.BF.PRB.OL.Rank4} + \text{L.ChMeas.MIMO.PRB.CL.Rank1} + \text{L.ChMeas.MIMO.PRB.CL.Rank2} + \text{L.ChMeas.MIMO.PRB.CL.Rank3} + \text{L.ChMeas.MIMO.PRB.CL.Rank4} + \text{L.ChMeas.MIMO.PRB.OL.Rank1} + \text{L.ChMeas.MIMO.PRB.OL.Rank2} + \text{L.ChMeas.MIMO.PRB.OL.Rank3} + \text{L.ChMeas.MIMO.PRB.OL.Rank4} + \text{L.ChMeas.MIMO.PRB.CL.Rank5} + \text{L.ChMeas.MIMO.PRB.CL.Rank6} + \text{L.ChMeas.MIMO.PRB.CL.Rank7} + \text{L.ChMeas.MIMO.PRB.CL.Rank8})}$
- Beamforming UEs using open-loop TM7, TM8, or TM9 (excluding UEs using closed-loop TM9). Open-loop TM9 UEs do not include UEs with rank 4 or above.

Proportion of PRBs for these UEs = $(L.ChMeas.BF.PR.B.OL.Rank1 + L.ChMeas.BF.PR.B.OL.Rank2 + L.ChMeas.BF.PR.B.OL.Rank3) / (L.ChMeas.BF.PR.B.OL.Rank1 + L.ChMeas.BF.PR.B.OL.Rank2 + L.ChMeas.BF.PR.B.OL.Rank3 + L.ChMeas.BF.PR.B.OL.Rank4 + L.ChMeas.MIMO.PR.B.CL.Rank1 + L.ChMeas.MIMO.PR.B.CL.Rank2 + L.ChMeas.MIMO.PR.B.OL.Rank1 + L.ChMeas.MIMO.PR.B.OL.Rank2 + L.ChMeas.MIMO.PR.B.CL.Rank3 + L.ChMeas.MIMO.PR.B.CL.Rank4 + L.ChMeas.MIMO.PR.B.OL.Rank3 + L.ChMeas.MIMO.PR.B.OL.Rank4)$

7.2.2 Impacts

The impacts between this function and other functions are described in [7.2.2 Impacts](#).

This function has the following impacts on LTE and NR network performance:

- This function increases the power consumed by the physical downlink shared channel (PDSCH), possibly increases the average MCS index, and reduces the downlink PRB usage.
 - LTE: Average MCS index = $\sum(n \times L.ChMeas.PDSCH.MCS.n) / \sum(L.ChMeas.PDSCH.MCS.n)$, where n ranges from 0 to 28 (for example, **L.ChMeas.PDSCH.MCS.1**). Downlink PRB usage = **L.ChMeas.PR.B.DL.Used.Avg/L.ChMeas.PR.B.DL.Avail** × 100%.
 - NR: Average MCS index = $\sum(n \times N.ChMeas.PDSCH.MCS.n) / \sum(N.ChMeas.PDSCH.MCS.n)$, where n ranges from 0 to 28 (for example, **N.ChMeas.PDSCH.MCS.1**). Downlink PRB usage = **(N.PR.B.DL.Used.Avg/N.PR.B.DL.Avail.Avg)** × 100%.
- The impact of this function on the downlink packet delay (LTE: **L.Traffic.DL.PktDelay.Time**; NR: **N.QoS.DL.PktDelaygNBDU.Time**) is related to services. When the tail packet percentage (the proportion of the last TTI before the downlink buffer is emptied) is low, the downlink packet delay decreases. When the tail packet percentage is high, the downlink packet delay increases.
- After this function is enabled, the average uplink and downlink UE throughputs may decrease for cells that constantly share out power.
 - LTE: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput**.
 - NR: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput (DU)**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput (DU)**.
- After this function is enabled, the IBLERs and RBLERs of cells involved in power sharing may slightly fluctuate. The IBLERs and RBLERs are calculated using the formulas listed in [Table 7-2](#).

Table 7-2 Formulas for calculating the IBLERs and RBLERs

LTE/ NR	Performance Indicator	Definition
LTE	Downlink IBLER	$\frac{(L.Traffic.DL.SCH.QPSK.ErrTB.Ibler + L.Traffic.DL.SCH.16QAM.ErrTB.Ibler + L.Traffic.DL.SCH.64QAM.ErrTB.Ibler + L.Traffic.DL.SCH.256QAM.ErrTB.Ibler)}{(L.Traffic.DL.SCH.QPSK.TB + L.Traffic.DL.SCH.16QAM.TB + L.Traffic.DL.SCH.64QAM.TB + L.Traffic.DL.SCH.256QAM.TB)} \times 100\%$
LTE	Downlink RBLER	$\frac{(L.Traffic.DL.SCH.QPSK.ErrTB.Rbler + L.Traffic.DL.SCH.16QAM.ErrTB.Rbler + L.Traffic.DL.SCH.64QAM.ErrTB.Rbler + L.Traffic.DL.SCH.256QAM.ErrTB.Rbler)}{(L.Traffic.DL.SCH.QPSK.TB + L.Traffic.DL.SCH.16QAM.TB + L.Traffic.DL.SCH.64QAM.TB + L.Traffic.DL.SCH.256QAM.TB)} \times 100\%$
NR	Downlink IBLER	$\frac{(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$
NR	Downlink RBLER	$\frac{(N.DL.SCH.QPSK.ErrTB.Rbler + N.DL.SCH.16QAM.ErrTB.Rbler + N.DL.SCH.64QAM.ErrTB.Rbler + N.DL.SCH.256QAM.ErrTB.Rbler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$

- After this function is enabled, the proportion of tail packets in the cell receiving shared power may increase.
 - LTE: Proportion of the last TTIs before the downlink buffer is empty = $(L.Thrp.Time.DL - L.Thrp.Time.DL.RmvLastTTI)/L.Thrp.Time.DL$
 - NR: Proportion of the last TTIs before the downlink buffer is empty = $(N.ThpTime.DL - N.ThpTime.DL.RmvLastSlot)/N.ThpTime.DL$
- After this function is enabled, a change in the control format indicator (CFI) may occur. In this case, the values of related indicators such as the number of available CCEs and CCE usage fluctuate.
- After this function is enabled, the values of the counters related to MU-MIMO pairing, such as **L.ChMeas.MassiveMIMO.MUBF.Avg.PairLayer**, **L.ChMeas.DL.Succ.Pair.TTI**, **N.ChMeas.MIMO.DL.Pair.Layer.Avg**, and **N.ChMeas.MIMO.DL.Pair.Ratio**, fluctuate.

- Function activation or deactivation involves the adjustment of system parameters, which causes service rate fluctuations for approximately 2 minutes and affects the cell throughput. Therefore, it is recommended that you enable this function when the traffic is light.
- When this function is enabled, the RF module power utilization increases. Base stations consume more power, but the total power consumption does not exceed the maximum power consumed when all cells work with a full load.
- This function requires information exchanges between cells. This may increase the CPU load of baseband processing units (**VS.BBUBoard.CPULoad.Mean**) by less than 5%.
- This function increases the instantaneous power used for PDSCH, which may increase interference to neighboring cells. It is recommended that this function be enabled in contiguous coverage areas. If the average cell throughput decreases, or if the interference to neighboring cells significantly increases after this function is enabled, you can decrease the maximum percentage of allocated power to counteract the negative impacts. (Setting the maximum percentage of allocated power to half of its recommended value is recommended.) If the gains are not noticeable, you can increase the value of the maximum percentage of allocated power to achieve higher function gains.
 - LTE: The maximum percentage of allocated power is specified by the **CellDynPowerSharing.MaxPowerSharingRate** parameter.
 - NR: The maximum percentage of allocated power is specified by the **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** parameter.
- It is recommended that this function be disabled before the downlink peak rate tests on an eNodeB or gNodeB.

7.3 Requirements

7.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE TDD	TDLCOFD-210611	Multi-band Power Adaptive Scheduling	LT4S00MBPAS0	per Cell
NR	FOFD-090100	Multi-band Power Adaptive Scheduling	NR0S00MBPA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

This function has no prerequisite functions on the LTE TDD side.

The prerequisite functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (<i>eRAN Feature Documentation</i>)	Description
Low-frequency NR TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	LTE and NR multi-band power adaptive scheduling requires NR multi-band power adaptive scheduling.

7.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions on the LTE TDD side are listed below.

RAT	Function Name	Function Switch	Reference (<i>eRAN Feature Documentation</i>)
LTE TDD	DL ICIC	ENodeBAlgoSwitch.DlIcicSwitch not set to DlIcicSwitch_OFF_ENUM	None

RAT	Function Name	Function Switch	Reference (<i>eRAN Feature Documentation</i>)
LTE TDD	CSPC	TddCspcMacroMicroSwitch option of the CspcAlgoPara.TddCspcAlgoSwitch parameter	<i>CSPC</i>
LTE TDD	Power adaptation for beamforming UEs	BFModeUserAdptPwrSwitch option of the CellAlgoSwitch.DlPcAlgoSwitch parameter	None
LTE TDD	Power optimization for beamforming UEs	BFModeUserPwrSwitch option of the CellAlgoSwitch.DlPcAlgoSwitch parameter	None
LTE TDD	CSPC	CellCspcPara.CellCspcSwitch	<i>CSPC</i>
LTE TDD	Enhanced coordinated scheduling based power control	CellCspcPara.CelleCspcSwitch	<i>Massive MIMO Enhancements (TDD)</i>
LTE TDD	PHICH inner-loop power control	PhichInnerLoopPcSwitch option of the CellAlgoSwitch.DlPcAlgoSwitch parameter	<i>Power Control</i>
LTE TDD	PDCCH spare power allocation optimization	CellPdcchAlgo.PdcchSparePowerAllocationStrage	<i>Physical Channel Resource Management</i>
LTE TDD	B-Trunc mode	eNodeBFunction.eNodeBMode set to BTRUNC_MODE	None
LTE TDD	Cell-level PDSCH power adjustment	CELL_LEVEL_PDSCH_POWER_ADJUST_SWITCH option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	None

RAT	Function Name	Function Switch	Reference (<i>eRAN Feature Documentation</i>)
LTE TDD	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_BAND_CARR_SW option of the ContigIntraBandCarr.ContigIntraBandCarrSw parameter	<i>Seamless Intra-Band Carrier Joining (TDD)</i>
LTE TDD	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>
LTE TDD	PDCCH SDMA ^a	MASSIVE_MIMO_PDCCH_SDMA_SW option of the CellPdcchAlgo.SplitBeamPdcchSdmaSw parameter	<i>Massive MIMO Basics (TDD)</i>
a: When LTE and NR multi-band power adaptive scheduling and PDCCH SDMA are both enabled, the CellPdcchAlgo.PdcchSdmaWeightType parameter cannot be set to POWER_SHARING_AMONG_3_BEAMS , POWER_SHARING_AMONG_4_BEAMS , or POWER_SHARING_AMONG_5_BEAMS in PDCCH SDMA.			

This function has no mutually exclusive functions on the NR side.

7.3.2.3 Function Impacts

The impacted functions on the LTE TDD side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Scalable bandwidth	None	<i>Cell Management</i>	10 MHz LTE cells that use default channel configurations seldom allocate power to NR cells as these LTE cells have only few power resources left after power reservation for paging channels, control channels, and traffic bursts.
LTE TDD	DL 256QAM	CellAlgoSwitch.DL256QamAlgoSwitch	<i>Modulation Schemes</i>	When UEs for which DL 256QAM has taken effect are scheduled in a TTI on the LTE side, LTE does not use the power allocated from NR.
LTE TDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDy nMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	LTE and NR multi-band power adaptive scheduling does not take effect when RF channel dynamic muting is in effect.
LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the CellPwrSavin gAlgo.PwrSavinAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Energy saving based on flexible frequency-domain scheduling is not performed on subframes where allocated power is used.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown . CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.Dyna micPowerSharingSwitch parameter deselected does not participate in LTE and NR multi-band power adaptive scheduling if it has entered the energy saving state. The cell will attempt to participate in power scheduling every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.Dyna micPowerSharingSwitch parameter is selected, power adaptive scheduling can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown . CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in LTE and NR multi-band power adaptive scheduling if it has entered the energy saving state. The cell will attempt to participate in power scheduling every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, power adaptive scheduling can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LTE TDD	Multi-carrier coordinated energy saving	CellShutdown . CellShutdo wnSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MUL_CARR_HIER_MA NUAL_SHUT DN , or ON_CARRIER_PRECISE_SH UTDOWN	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OP T_SW option of the CellDynPower Sharing.Dyna micPowerShar ingSwitch parameter deselected does not participate in LTE and NR multi-band power adaptive scheduling if it has entered the energy saving state. The cell will attempt to participate in power scheduling every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OP T_SW option of the CellDynPower Sharing.Dyna micPowerShar ingSwitch parameter is selected, power adaptive scheduling can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LTE TDD	Automatic cell shutdown	CellAutoShut down.CellAut oShutdownS witch	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OP T_SW option of the CellDynPower Sharing.Dyna micPowerShar ingSwitch parameter deselected does not participate in LTE and NR multi-band power adaptive scheduling if it has entered the energy saving state. The cell will attempt to participate in power scheduling every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OP T_SW option of the CellDynPower Sharing.Dyna micPowerShar ingSwitch parameter is selected, power adaptive scheduling can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavin gAlgo.Carrier DynShutdow nPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPower Sharing.Dyna micPowerShar ingSwitch parameter deselected does not participate in LTE and NR multi-band power adaptive scheduling if it has entered the energy saving state. The cell will attempt to participate in power scheduling every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPower Sharing.Dyna micPowerShar ingSwitch parameter is selected, power adaptive scheduling can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Cells that have entered the energy saving state do not participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE TDD	Enhanced channel power forming for beamforming	BfEnhChnPwrFormSwitch option of the CellAlgoSwitch.DIPcAlgoSwitch parameter	None	When both enhanced channel power forming for beamforming and LTE and NR multi-band power adaptive scheduling are enabled, the power aggregation amount for the former function decreases. This may result in a higher downlink block error rate (BLER) and a lower average downlink UE throughput than those when only enhanced channel power forming for beamforming is enabled.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over LTE and NR multi-band power adaptive scheduling when both functions are enabled. LTE and NR multi-band power adaptive scheduling does not take effect when RF module fast deep dormancy is in effect.

The impacted functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	When UEs for which DL 256QAM has taken effect are scheduled in a slot on the NR side, NR does not use the power allocated from LTE.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULING_SW option of the NRDUCellAlg oSwitch. PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When LTE and NR multi-band power adaptive scheduling and power saving scheduling are both enabled, the effects of power saving scheduling are affected. It is recommended that LTE and NR multi-band power adaptive scheduling not be enabled with this function when the traffic of LTE and NR cells is light.
Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlg oSwitch. PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When LTE and NR multi-band power adaptive scheduling takes effect together with energy saving based on flexible frequency-domain scheduling, the benefits of both functions decrease.
Low-frequency NR TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlg oSwitch. CompSwitch parameter	<i>CoMP</i>	When LTE and NR multi-band power adaptive scheduling and intra-base-station DL CoMP are both enabled, the benefits of power adaptive scheduling may decrease on the NR side.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCol labServ.Multi CellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When LTE and NR multi-band power adaptive scheduling and Co-MM are both enabled, the benefits of power adaptive scheduling may decrease on the NR side.
Low-frequency NR TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	When LTE and NR multi-band power adaptive scheduling and coordinated interference management are both enabled, the benefits of power adaptive scheduling may decrease on the NR side.
Low-frequency NR TDD	Power reduction for CCUs	None	<i>Power Control</i>	When LTE and NR multi-band power adaptive scheduling and power reduction for CCUs are both enabled, the benefits of power adaptive scheduling may decrease on the NR side.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW or INTRF_COORD_ALWAYS_WORKING_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	When LTE and NR multi-band power adaptive scheduling and interference coordination between public and dedicated networks are both enabled, the benefits of power adaptive scheduling may decrease on the NR side.
Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	A carrier exits the power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, system parameters are adjusted, resulting in fluctuations in SSB coverage, CSI-RS coverage, PDCCH coverage, and control message coverage. Consequently, the access success rate, handover success rate, service drop rate, and paging success rate fluctuate, and service fluctuation lasts for approximately 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	A carrier exits the power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, system parameters are adjusted, resulting in fluctuations in SSB coverage, CSI-RS coverage, PDCCH coverage, and control message coverage. Consequently, the access success rate, handover success rate, service drop rate, and paging success rate fluctuate, and service fluctuation lasts for approximately 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Cells that have entered the energy saving state do not participate in power adaptive scheduling. When a carrier enters or exits the energy saving state, system parameters are adjusted, resulting in fluctuations in SSB coverage, CSI-RS coverage, PDCCH coverage, and control message coverage. Consequently, the access success rate, handover success rate, service drop rate, and paging success rate fluctuate, and service fluctuation lasts for approximately 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	TTI-level channel shutdown	NRDUCellPowerSaving.C o mChnDynSh utdownSwitc h parameter and TTI_RF_SHUT DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwit ch parameter	<i>Energy Conservation and Emission Reduction</i>	When LTE and NR multi-band power adaptive scheduling takes effect together with TTI-level channel shutdown, the energy saving benefits of TTI-level channel shutdown decrease and downlink UE throughput may decrease.
Low-frequency NR TDD	PDCCH aggregation level reduction	PDCCH_AGG_LVL_COMPR_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When LTE and NR multi-band power adaptive scheduling and PDCCH aggregation level reduction are both enabled, only LTE and NR multi-band power adaptive scheduling takes effect.
Low-frequency NR TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDIA mc.DlModulationSchemeS w parameter	<i>Modulation Schemes</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where LTE and NR multi-band power adaptive scheduling takes effect decreases.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	NR low power consumption mode	LOW_PWR_MODE_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Only low power consumption mode takes effect when low power consumption mode and LTE and NR multi-band power adaptive scheduling are both enabled and a customized backup power saving policy is used in low power consumption mode.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	Maximum SSB power offset	NRDUCellTrpBeam.MaxSsbPwrOffset	<i>Power Control</i>	When the NRDUCellTrpBeam.MaxSsbPwrOffset parameter is set to a value greater than 3 dB and LTE and NR multi-band power adaptive scheduling is in effect: • If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is deselected, the SSB power aggregation capability decreases. • If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is selected, the SSB power aggregation capability does not decrease. This is because, with this option selected, LTE and NR multi-band power

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				adaptive scheduling will not be performed (and will not provide gains) in slots with SSB transmission.
Low-frequency NR TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	A carrier exits the power adaptive scheduling group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, system parameters are adjusted, resulting in fluctuations in SSB coverage, CSI-RS coverage, PDCCH coverage, and control message coverage. Consequently, the access success rate, handover success rate, service drop rate, and paging success rate fluctuate, and service fluctuation lasts for approximately 2 minutes.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

LTE TDD: 3900 and 5900 series base stations (macro base stations)

NR: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

The main control board must be a UMPTE or a later board, and the baseband processing unit must be a UBBPfw, a UBBPg, or a later TDD-capable board.

RF Modules

AAU5368md (whose SBOM is not 02315DBT) and AAU5665md

7.3.4 Cells

Requirements on the LTE TDD side:

- **Cell.TxRxMode** in a **Cell** MO must be set to **64T64R** or **32T32R**.
- **Cell.MultiRruCellMode** is not set to **ASSISTED_MASSIVE_MIMO_CELL**.
- **PHICHCfg.PhichDuration** is not set to **EXTENDED**.
- **Cell.MultiCellShareMode** is set to **NORMAL**.

There are no requirements on the NR side.

7.3.5 Networking

LTE cells must work on the same frequency band. Otherwise, this function cannot be enabled. The AAU5368md is used as an example. It can work on the 2.3 GHz and 3.5 GHz frequency bands. All LTE cells served by this module must work on the 2.3 GHz or 3.5 GHz frequency band to enable this function.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

This feature takes effect only after parameters are configured on both the eNodeB and gNodeB sides. [Table 7-3](#) and [Table 7-4](#) describe the parameters used for function activation on the eNodeB and gNodeB sides, respectively.

Table 7-3 eNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	CellDynPowerSharing.DynamicPowerSharingSwitch	Select the LNR_MULTI_BAND_PWR_ADPT_SCH_SW option if LTE and NR multi-band power adaptive scheduling is required.
Max Power Sharing Rate	CellDynPowerSharing.MaxPowerSharingRate	Use the recommended value.
Multi-Band Power Adapt Scheduling Priority	CellDynPowerSharing.MultiBandPwrAdaptSchPri	Use the recommended value.

Table 7-4 gNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	NRDUCellAlgoSwitch.DynPowerSharingSwitch	Select the LNR_MULTI_BAND_PWR_ADPT_SCH_SW option if LTE and NR multi-band power adaptive scheduling is required.
Max Power Sharing Rate	NRDUCellTrpDynPwrShr.MaxPowerSharingRate	Use the recommended value.
Multi-Band Power Adapt Scheduling Priority	NRDUCellTrpDynPwrShr.MultiBandPwrAdaptSchPri	Use the recommended value.

7.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//On the eNodeB side
MOD CELLDYNPOWERSHARING: LocalCellId=0,
```

```
DynamicPowerSharingSwitch=LNR_MULTI_BAND_PWR_ADPT_SCH_SW-1;
MOD CELLDYNPOWERSHARING: LocalCellId=0, MaxPowerSharingRate=50, MultiBandPwrAdaptSchPri=0;
//On the gNodeB side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
DynPowerSharingSwitch=LNR_MULTI_BAND_PWR_ADPT_SCH_SW-1;
MOD NRDUCELLTRPDYNPWRSHR: NrDuCellTrpId=0, MaxPowerSharingRate=100,
MultiBandPwrAdaptSchPri=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//On the eNodeB side
MOD CELLDYNPOWERSHARING: LocalCellId=0,
DynamicPowerSharingSwitch=LNR_MULTI_BAND_PWR_ADPT_SCH_SW-0;
//On the gNodeB side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
DynPowerSharingSwitch=LNR_MULTI_BAND_PWR_ADPT_SCH_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

Perform the following steps to check whether this function has taken effect.

1. Check whether this function has taken effect in LTE cells.
Run the eNodeB MML command **DSP CELLDYNPWRSHASTATUS**. If the value of **LNR Multi-Band Power Adapt Scheduling Status** is **EFFECTIVE**, this function has taken effect in LTE cells.
2. Check whether this function has taken effect in NR cells.
Run the gNodeB MML command **DSP NRDUCELLTRPDYNPWRSHR**. If the value of **LNR Multi-Band Adapt Pwr Sch Valid Status** is **Enabled**, this function has taken effect in NR cells.
3. Monitor the values of the counters listed in the tables below on the MAE. If the counter values are greater than 0, this function has taken effect. If both the counter values are 0, this function has not taken effect.

Table 7-5 eNodeB counters

Counter ID	Counter Name
1526746987	L.PwrShare.PwrOut.TTI.Num
1526746986	L.PwrShare.PwrIn.TTI.Num

Table 7-6 gNodeB counters

Counter ID	Counter Name
1911823028	N.PwrShare.PwrOut.Slot.Num
1911823027	N.PwrShare.PwrIn.Slot.Num

 NOTE

- After LTE and NR multi-band power adaptive scheduling is activated, LTE and NR cells will negotiate the configuration of dynamic power adaptive scheduling. Generally, the negotiation can be completed within 5 minutes after the last activation step. Therefore, it is recommended that the activation status be queried 5 minutes after the activation procedure is complete.
- Intra-RAT power sharing is preferentially performed in an LTE and NR multi-band power adaptive scheduling group. If there is still surplus power after intra-RAT power sharing, inter-RAT power scheduling can be performed. The performance counters related to power scheduling measure the actual time when each cell has its power allocated or receives allocated power, without distinguishing between intra-RAT and inter-RAT power scheduling.
- LTE measures the number of TTIs in which a cell has its power allocated and the number of TTIs in which a cell receives allocated power in each 1-ms TTI. NR measures the number of slots in which a cell has its power allocated and the number of slots in which a cell receives allocated power.
- The **L.PwrShare.PwrIn.TTI.Num** counter indicates the number of TTIs within which the cell uses received allocated power. As power scheduling uses the estimation results from the previous TTI, the estimated number of TTIs within which the cell has received allocated power might be incorrect.
- The **N.PwrShare.PwrIn.Slot.Num** counter indicates the number of slots within which the cell uses received allocated power. As power scheduling uses the estimation results from the previous millisecond, the estimated number of slots within which the cell has received allocated power might be incorrect.
- When a slot of an NR cell is counted as a slot within which the cell receives allocated power, the actual power used in the slot is generally greater than the configured static cell power. However, there are also exceptions. When PDSCH power aggregation is disabled for an NR cell, the actual power used in the slot is less than the configured static cell power if there are remaining RBs in the slot and the power of the remaining RBs is greater than the increased power of beneficiary UEs.
- When the power of a single cell is shared with one or multiple cells in an LTE and NR multi-band power adaptive scheduling group, the number of TTIs in which a cell uses received allocated power is counted for each cell. Therefore, the number of TTIs in which a cell has its power scheduled for other cells differs from the number of TTIs in which a cell uses received allocated power.

7.4.3 Network Monitoring

LTE Network Monitoring

When event A3 can be reported, determine whether this feature has taken effect based on the throughput of CEUs, which can be calculated based on the information contained in event A3 reports. The detailed steps are as follows:

- Step 1** One week before feature activation, select the **BasedA3EdgeUserSwitch** option of the **CellCounterParaGroup.CellCounterAlgoSwitch** parameter, and set **CellCounterParaGroup.EdgeUserA3Offset** to **-12**.

 NOTE

The average downlink throughput of CEUs can be measured only when the **BasedA3EdgeUserSwitch** option of the **CellCounterParaGroup.CellCounterAlgoSwitch** parameter is selected.

Step 2 Observe the values of the following counters to monitor feature gains.

- Average downlink UE throughput = $(L.Thrp.bits.DL - L.Thrp.bits.DL.LastTTI) / L.Thrp.Time.DL.RmvLastTTI$
- Average downlink throughput of CEUs = $(L.Thrp.bits.DL.BorderUE - L.Thrp.bits.DL.LastTTI.BorderUE) / L.Thrp.Time.DL.RmvLastTTI.BorderUE$

----End

When event A3 cannot be reported, determine whether this feature has taken effect based on the cumulative distribution function (CDF) curve of the throughput.

Create the CDF curve based on values of counters listed in [Table 7-7](#). If the throughput corresponding value 5% or 10% on the CDF curve increases, this feature has taken effect.

Table 7-7 LTE counters

Counter ID	Counter Name
1526729056 to 1526729065	L.Thrp.DL.BitRate.Samp.Index0 to L.Thrp.DL.BitRate.Samp.Index9

NR Network Monitoring

Step 1 Select the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter one week before feature activation.

 NOTE

The average downlink throughput of UEs in overlapping areas can be measured only when the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter is selected.

Step 2 Observe the values of the following counters to monitor feature gains.

- Average downlink UE throughput = $(N.ThpVol.DL - N.ThpVol.DL.LastSlot) / N.ThpTime.DL.RmvLastSlot$
- Average downlink throughput of UEs in overlapping areas = $(N.ThpVol.DL.CellOverlap - N.ThpVol.DL.LastSlot.CellOverlap) / N.ThpTime.DL.RmvLastSlot.CellOverlap$

----End

8 High Static PSD Ratios Between LTE TDD and NR

8.1 Principles

When LTE and NR networks coexist and the traffic volume on LTE networks is still high, the transmit power of LTE networks needs to be maintained or increased to ensure the LTE network coverage.

This function allows operators to adjust RF configurations so that LTE cells can have a higher static power, achieving a high static power spectral density (PSD) ratio between LTE and NR. This function is enabled by selecting the **SPEC_ENH_LOOSE_PSD_LIMIT** option of the LTE parameter **RRU.CUSTOMEDRFSPECSW** and selecting the **SPEC_ENH_LOOSE_PSD_LIMIT** option of the NR parameter **RRU.CUSTOMEDRFSPECSW**.

8.2 Network Analysis

8.2.1 Benefits

This function implements a high static PSD ratio between LTE and NR. It allows a higher static power to be configured for LTE cells, improving the coverage of LTE cells. It is recommended that this function be enabled in scenarios where the LTE cell coverage is limited, such as suburban or rural areas.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

The increase in the LTE static PSD requires BBU-RRU joint clipping on the LTE side. As a result, the CQI may slightly decrease and the RRC connection setup delay may slightly increase.

8.3 Requirements

8.3.1 Licenses

There are no license requirements for basic functions.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

This feature involves the cooperation between LTE and NR. Therefore, it is recommended that LTE and NR be upgraded to the same version.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

None

8.3.2.3 Function Impacts

There are no function impacts on the LTE TDD side.

The function impacts on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Low-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When AAUs are used, cells that have entered the energy saving state do not support high static PSD ratios between LTE and NR. When a carrier enters or exits the energy saving state, system parameters are adjusted, resulting in fluctuations in SSB coverage, CSI-RS coverage, PDCCH coverage, and control message coverage. Consequently, the access success rate, handover success rate, service drop rate, and paging success rate fluctuate, and service fluctuation lasts for approximately 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Only low power consumption mode takes effect when both low power consumption mode and high static PSD ratios between LTE TDD and NR are enabled and a customized backup power saving policy is used in low power consumption mode.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- LTE: 3900 and 5900 series base stations (macro base stations). 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.
- NR: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

- Main control board: UMPT
- Baseband processing unit: UBBPd and later

RF Modules

Only the RRU5258 (SBOM: 02312AFL, 02312VPV, 02312AFM, 02312QFK, 02312AFN, 02312KUR, 02312DAB, 02312QFS, 02312PHB, and 02312QFQ), RRU5818 (SBOM: 02313BDD, 02313BDE, 02313NGM, and 02312LWK), and RRU5872 support this function.

Cells

When RRUs are used, this function can only be enabled between one or two 20 MHz LTE cells and an NR cell with a 20–100 MHz bandwidth. These cells must be configured with contiguous spectrum. In addition, the cells cannot be combined-RRU cells.

If the preceding requirements on cells are not met, the cells fail to be activated after this function is enabled.

If the preceding requirements on boards or RF modules are not met, this function does not take effect after it is enabled, but the cell status is normal.

NOTE

Before enabling this feature, ensure that sufficient power licenses (TD1SLPT10W00) are available.

8.3.4 Others

None

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

This function takes effect only after parameters are configured on both the eNodeB and gNodeB sides. [Table 8-1](#) and [Table 8-2](#) describe the parameters used for function activation on the eNodeB and gNodeB sides, respectively.

Table 8-1 eNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
RF Specifications Customization Switch	RRU.CUSTOMEDRFSP ECSW	Select the SPEC_ENH_LOOSE_PSD_LIMIT option.

Table 8-2 gNodeB parameters used for activation

Parameter Name	Parameter ID	Setting Notes
RF Specifications Customization Switch	RRU.CUSTOMEDRFSP ECSW	Select the SPEC_ENH_LOOSE_PSD_LIMIT option.

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//On the eNodeB side  
MOD RRU: CN=0, SRN=60, SN=0, CUSTOMEDRFSPECSW=SPEC_ENH_LOOSE_PSD_LIMIT-1,  
FORCEEXECUTE=YES;  
//On the gNodeB side  
MOD RRU: CN=0, SRN=60, SN=0, CUSTOMEDRFSPECSW=SPEC_ENH_LOOSE_PSD_LIMIT-1,  
FORCEEXECUTE=YES;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//On the eNodeB side  
MOD RRU: CN=0, SRN=60, SN=0, CUSTOMEDRFSPECSW=SPEC_ENH_LOOSE_PSD_LIMIT-0,  
FORCEEXECUTE=YES;  
//On the gNodeB side  
MOD RRU: CN=0, SRN=60, SN=0, CUSTOMEDRFSPECSW=SPEC_ENH_LOOSE_PSD_LIMIT-0,  
FORCEEXECUTE=YES;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

After this function is enabled, the static power of LTE cells can be increased. After the LTE cell is successfully activated, run the **LST PDSCHCFG** command. If the value of **Reference signal power(0.1dBm)** in the command output is the same as the configured static power, this function has taken effect.

8.4.3 Network Monitoring

This function ensures basic network performance and does not require network monitoring.

9 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- 3GPP TS 38.104: "NR;Base Station (BS) radio transmission and reception"
- eRAN feature parameter description documents
 - *Dynamic Power Sharing Between LTE Carriers*
 - *Cell Management*
 - *Physical Channel Resource Management*
 - *Massive MIMO Basics (TDD)*
 - *Energy Conservation and Emission Reduction*
 - *Carrier Aggregation*
 - *On-Demand TX Power Allocation Under EME*
 - *Multi-RAT Coordinated Energy Saving*
 - *LTE FDD and NR Dynamic Spectrum Sharing*
 - *Smart 8T8R (FDD)*
 - *Massive MIMO Enhancements (FDD)*
 - *Soft Split Resource Duplex (TDD)*
 - *Intelligent Multi-Beam of 8T8R (TDD)*
 - *Seamless Intra-Band Carrier Joining (TDD)*
 - *CSPC*
 - *Green Site*
 - *Interference Detection and Suppression*
 - *SFN*
 - *Massive MIMO Enhancements (TDD)*
 - *License Control Item Lists (FDD)*
 - *License Control Item Lists (CIoT)*
 - *License Control Item Lists (TDD)*
 - *Power Control*
 - *GSM and LTE Dynamic Power Sharing*
 - *Relay*
 - *Seamless Intra-Band Carrier Joining (FDD)*

- *Virtual 4T4R (FDD)*
 - *NB-IoT Basics (FDD)*
 - *eMTC*
 - *Compact Bandwidth (FDD)*
 - *LTE Flexible Spectrum (FDD)*
 - *SFN*
- 5G RAN feature parameter description documents
 - *Modulation Schemes*
 - *Energy Conservation and Emission Reduction*
 - *High Speed Mobility*
 - *CoMP*
 - *Hyper Cell*
 - *Cell Combination*
 - *Power Control*
 - *Coordinated Interference Management (Low-Frequency TDD)*
 - *Co-MM (Low-Frequency TDD)*
 - *LTE FDD and NR Dynamic Spectrum Sharing*
 - *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
 - *NR Inter-Carrier Dynamic Power Allocation*
 - *Massive MIMO iBeam (Low-Frequency TDD)*
 - *Downlink Scheduling*
 - *Distributed Antenna Solutions (FDD)*
 - *Fusion Cell (Low-Frequency TDD)*
 - *Channel Management*
 - *Turbo Coverage (FDD)*
 - *Distributed Antenna Solutions (Low-Frequency TDD)*
 - *Multi-RAT Coordinated Energy Saving*
 - *Standards Compliance*
 - *Interference Avoidance*
 - *License Control Item Lists*